

Table A2. Binomial distribution

$$F(x) = \mathbf{P}\{X \leq x\} = \sum_{k=0}^x \binom{n}{k} p^k (1-p)^{n-k}$$

n	x	p																		
		.050	.100	.150	.200	.250	.300	.350	.400	.450	.500	.550	.600	.650	.700	.750	.800	.850	.900	.950
1	0	.950	.900	.850	.800	.750	.700	.650	.600	.550	.500	.450	.400	.350	.300	.250	.200	.150	.100	.050
2	0	.903	.810	.723	.640	.563	.490	.423	.360	.303	.250	.203	.160	.123	.090	.063	.040	.023	.010	.003
	1	.998	.990	.978	.960	.938	.910	.878	.840	.798	.750	.698	.640	.578	.510	.438	.360	.278	.190	.098
3	0	.857	.729	.614	.512	.422	.343	.275	.216	.166	.125	.091	.064	.043	.027	.016	.008	.003	.001	.000
	1	.993	.972	.939	.896	.844	.784	.718	.648	.575	.500	.425	.352	.282	.216	.156	.104	.061	.028	.007
	2	1.0	.999	.997	.992	.984	.973	.957	.936	.909	.875	.834	.784	.725	.657	.578	.488	.386	.271	.143
4	0	.815	.656	.522	.410	.316	.240	.179	.130	.092	.063	.041	.026	.015	.008	.004	.002	.001	.000	.000
	1	.986	.948	.890	.819	.738	.652	.563	.475	.391	.313	.241	.179	.126	.084	.051	.027	.012	.004	.000
	2	1.0	.996	.988	.973	.949	.916	.874	.821	.759	.688	.609	.525	.437	.348	.262	.181	.110	.052	.014
	3	1.0	1.0	.999	.998	.996	.992	.985	.974	.959	.938	.908	.870	.821	.760	.684	.590	.478	.344	.185
5	0	.774	.590	.444	.328	.237	.168	.116	.078	.050	.031	.018	.010	.005	.002	.001	.000	.000	.000	.000
	1	.977	.919	.835	.737	.633	.528	.428	.337	.256	.188	.131	.087	.054	.031	.016	.007	.002	.000	.000
	2	.999	.991	.973	.942	.896	.837	.765	.683	.593	.500	.407	.317	.235	.163	.104	.058	.027	.009	.001
	3	1.0	1.0	.998	.993	.984	.969	.946	.913	.869	.813	.744	.663	.572	.472	.367	.263	.165	.081	.023
	4	1.0	1.0	1.0	1.0	.999	.998	.995	.990	.982	.969	.950	.922	.884	.832	.763	.672	.556	.410	.226
6	0	.735	.531	.377	.262	.178	.118	.075	.047	.028	.016	.008	.004	.002	.001	.000	.000	.000	.000	.000
	1	.967	.886	.776	.655	.534	.420	.319	.233	.164	.109	.069	.041	.022	.011	.005	.002	.000	.000	.000
	2	.998	.984	.953	.901	.831	.744	.647	.544	.442	.344	.255	.179	.117	.070	.038	.017	.006	.001	.000
	3	1.0	.999	.994	.983	.962	.930	.883	.821	.745	.656	.558	.456	.353	.256	.169	.099	.047	.016	.002
	4	1.0	1.0	1.0	.998	.995	.989	.978	.959	.931	.891	.836	.767	.681	.580	.466	.345	.224	.114	.033
	5	1.0	1.0	1.0	1.0	1.0	.999	.998	.996	.992	.984	.972	.953	.925	.882	.822	.738	.623	.469	.265
7	0	.698	.478	.321	.210	.133	.082	.049	.028	.015	.008	.004	.002	.001	.000	.000	.000	.000	.000	.000
	1	.956	.850	.717	.577	.445	.329	.234	.159	.102	.063	.036	.019	.009	.004	.001	.000	.000	.000	.000
	2	.996	.974	.926	.852	.756	.647	.532	.420	.316	.227	.153	.096	.056	.029	.013	.005	.001	.000	.000
	3	1.0	.997	.988	.967	.929	.874	.800	.710	.608	.500	.392	.290	.200	.126	.071	.033	.012	.003	.000
	4	1.0	1.0	.999	.995	.987	.971	.944	.904	.847	.773	.684	.580	.468	.353	.244	.148	.074	.026	.004
	5	1.0	1.0	1.0	1.0	.999	.996	.991	.981	.964	.938	.898	.841	.766	.671	.555	.423	.283	.150	.044
	6	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.996	.992	.985	.972	.951	.918	.867	.790	.679	.522	.302
8	0	.663	.430	.272	.168	.100	.058	.032	.017	.008	.004	.002	.001	.000	.000	.000	.000	.000	.000	.000
	1	.943	.813	.657	.503	.367	.255	.169	.106	.063	.035	.018	.009	.004	.001	.000	.000	.000	.000	.000
	2	.994	.962	.895	.797	.679	.552	.428	.315	.220	.145	.088	.050	.025	.011	.004	.001	.000	.000	.000
	3	1.0	.995	.979	.944	.886	.806	.706	.594	.477	.363	.260	.174	.106	.058	.027	.010	.003	.000	.000
	4	1.0	1.0	.997	.990	.973	.942	.894	.826	.740	.637	.523	.406	.294	.194	.114	.056	.021	.005	.000
	5	1.0	1.0	1.0	1.0	.999	.996	.989	.975	.950	.912	.855	.780	.685	.572	.448	.321	.203	.105	.038
	6	1.0	1.0	1.0	1.0	1.0	.999	.996	.991	.982	.965	.937	.894	.831	.745	.633	.497	.343	.187	.057
	7	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.996	.992	.983	.968	.942	.900	.832	.728	.570	.337
9	0	.630	.387	.232	.134	.075	.040	.021	.010	.005	.002	.001	.000	.000	.000	.000	.000	.000	.000	.000
	1	.929	.775	.599	.436	.300	.196	.121	.071	.039	.020	.009	.004	.001	.000	.000	.000	.000	.000	.000
	2	.992	.947	.859	.738	.601	.463	.337	.232	.150	.090	.050	.025	.011	.004	.001	.000	.000	.000	.000
	3	.999	.992	.966	.914	.834	.730	.609	.483	.361	.254	.166	.099	.054	.025	.010	.003	.001	.000	.000
	4	1.0	.999	.994	.980	.951	.901	.828	.733	.621	.500	.379	.267	.172	.099	.049	.020	.006	.001	.000
	5	1.0	1.0	.999	.997	.990	.975	.946	.901	.834	.746	.639	.517	.391	.270	.166	.086	.034	.008	.001
	6	1.0	1.0	1.0	1.0	.999	.996	.989	.975	.950	.910	.850	.768	.663	.537	.399	.262	.141	.053	.008
	7	1.0	1.0	1.0	1.0	1.0	1.0	.999	.996	.991	.980	.961	.929	.879	.804	.700	.564	.401	.225	.071
	8	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.995	.990	.979	.960	.925	.866	.768	.613	.370
10	0	.599	.349	.197	.107	.056	.028	.013	.006	.003	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000
	1	.914	.736	.544	.376	.244	.149	.086	.046	.023	.011	.005	.002	.001	.000	.000	.000	.000	.000	.000
	2	.988	.930	.820	.678	.526	.383	.262	.167	.100	.055	.027	.012	.005	.002	.000	.000	.000	.000	.000
	3	.999	.987	.950	.879	.776	.650	.514	.382	.266	.172	.102	.055	.026	.011	.004	.001	.000	.000	.000
	4	1.0	.998	.990	.967	.922	.850	.751	.633	.504	.377	.262	.166	.095	.047	.020	.006	.001	.000	.000
	5	1.0	1.0	.999	.994	.980	.953	.905	.834	.738	.623	.496	.367	.249	.150	.078	.033	.010	.002	.000
	6	1.0	1.0	1.0	.999	.996	.989	.974	.945	.898	.828	.734	.618	.486	.350	.224	.121	.050	.013	.001
	7	1.0	1.0	1.0	1.0	1.0	.998	.995	.988	.973	.945	.900	.833	.738	.617	.474	.322	.180	.070	.012
	8	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.995	.989	.977	.954	.914	.851	.756	.624	.456	.264	.086
	9	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.997	.994	.987	.972	.944	.893	.803	.651	.401

Table A2, continued. Binomial distribution

n	x	p																		
		.050	.100	.150	.200	.250	.300	.350	.400	.450	.500	.550	.600	.650	.700	.750	.800	.850	.900	.950
11	0	.569	.314	.167	.086	.042	.020	.009	.004	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	1	.898	.697	.492	.322	.197	.113	.061	.030	.014	.006	.002	.001	.000	.000	.000	.000	.000	.000	.000
	2	.985	.910	.779	.617	.455	.313	.200	.119	.065	.033	.015	.006	.002	.001	.000	.000	.000	.000	.000
	3	.998	.981	.931	.839	.713	.570	.426	.296	.191	.113	.061	.029	.012	.004	.001	.000	.000	.000	.000
	4	1.0	.997	.984	.950	.885	.790	.668	.533	.397	.274	.174	.099	.050	.022	.008	.002	.000	.000	.000
	5	1.0	1.0	.997	.988	.966	.922	.851	.753	.633	.500	.367	.247	.149	.078	.034	.012	.003	.000	.000
	6	1.0	1.0	1.0	.998	.992	.978	.950	.901	.826	.726	.603	.467	.332	.210	.115	.050	.016	.003	.000
	7	1.0	1.0	1.0	1.0	.999	.996	.988	.971	.939	.887	.809	.704	.574	.430	.287	.161	.069	.019	.002
	8	1.0	1.0	1.0	1.0	1.0	.999	.998	.994	.985	.967	.935	.881	.800	.687	.545	.383	.221	.090	.015
	9	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.994	.986	.970	.939	.887	.803	.678	.508	.303	.102
10	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.996	.991	.980	.958	.914	.833	.686	.431	
12	0	.540	.282	.142	.069	.032	.014	.006	.002	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	1	.882	.659	.443	.275	.158	.085	.042	.020	.008	.003	.001	.000	.000	.000	.000	.000	.000	.000	.000
	2	.980	.889	.736	.558	.391	.253	.151	.083	.042	.019	.008	.003	.001	.000	.000	.000	.000	.000	.000
	3	.998	.974	.908	.795	.649	.493	.347	.225	.134	.073	.036	.015	.006	.002	.000	.000	.000	.000	.000
	4	1.0	.996	.976	.927	.842	.724	.583	.438	.304	.194	.112	.057	.026	.009	.003	.001	.000	.000	.000
	5	1.0	.999	.995	.981	.946	.882	.787	.665	.527	.387	.261	.158	.085	.039	.014	.004	.001	.000	.000
	6	1.0	1.0	.999	.996	.986	.961	.915	.842	.739	.613	.473	.335	.213	.118	.054	.019	.005	.001	.000
	7	1.0	1.0	1.0	.999	.997	.991	.974	.943	.888	.806	.696	.562	.417	.276	.158	.073	.024	.004	.000
	8	1.0	1.0	1.0	1.0	1.0	.998	.994	.985	.964	.927	.866	.775	.653	.507	.351	.205	.092	.026	.002
	9	1.0	1.0	1.0	1.0	1.0	1.0	.999	.997	.992	.981	.958	.917	.849	.747	.609	.442	.264	.111	.020
	10	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.997	.992	.980	.958	.915	.842	.725	.557	.341	.118
	11	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.994	.986	.968	.931	.858	.718	.460
13	0	.513	.254	.121	.055	.024	.010	.004	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	1	.865	.621	.398	.234	.127	.064	.030	.013	.005	.002	.001	.000	.000	.000	.000	.000	.000	.000	.000
	2	.975	.866	.692	.502	.333	.202	.113	.058	.027	.011	.004	.001	.000	.000	.000	.000	.000	.000	.000
	3	.997	.966	.882	.747	.584	.421	.278	.169	.093	.046	.020	.008	.003	.001	.000	.000	.000	.000	.000
	4	1.0	.994	.966	.901	.794	.654	.501	.353	.228	.133	.070	.032	.013	.004	.001	.000	.000	.000	.000
	5	1.0	.999	.992	.970	.920	.835	.716	.574	.427	.291	.179	.098	.046	.018	.006	.001	.000	.000	.000
	6	1.0	1.0	.999	.993	.976	.938	.871	.771	.644	.500	.356	.229	.129	.062	.024	.007	.001	.000	.000
	7	1.0	1.0	1.0	.999	.994	.982	.954	.902	.821	.709	.573	.426	.284	.165	.080	.030	.008	.001	.000
	8	1.0	1.0	1.0	1.0	.999	.996	.987	.968	.930	.867	.772	.647	.499	.346	.206	.099	.034	.006	.000
	9	1.0	1.0	1.0	1.0	1.0	.999	.997	.992	.980	.954	.907	.831	.722	.579	.416	.253	.118	.034	.003
	10	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.996	.989	.973	.942	.887	.798	.667	.498	.308	.134	.025
	11	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.995	.987	.970	.936	.873	.766	.602	.379	.135
	12	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.996	.990	.976	.945	.879	.746	.487
14	0	.488	.229	.103	.044	.018	.007	.002	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	1	.847	.585	.357	.198	.101	.047	.021	.008	.003	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000
	2	.970	.842	.648	.448	.281	.161	.084	.040	.017	.006	.002	.001	.000	.000	.000	.000	.000	.000	.000
	3	.996	.956	.853	.698	.521	.355	.220	.124	.063	.029	.011	.004	.001	.000	.000	.000	.000	.000	.000
	4	1.0	.991	.953	.870	.742	.584	.423	.279	.167	.090	.043	.018	.006	.002	.000	.000	.000	.000	.000
	5	1.0	.999	.988	.956	.888	.781	.641	.486	.337	.212	.119	.058	.024	.008	.002	.000	.000	.000	.000
	6	1.0	1.0	.998	.988	.962	.907	.816	.692	.546	.395	.259	.150	.075	.031	.010	.002	.000	.000	.000
	7	1.0	1.0	1.0	.998	.990	.969	.925	.850	.741	.605	.454	.308	.184	.093	.038	.012	.002	.000	.000
	8	1.0	1.0	1.0	1.0	.998	.992	.976	.942	.881	.788	.663	.514	.359	.219	.112	.044	.012	.001	.000
	9	1.0	1.0	1.0	1.0	1.0	.998	.994	.982	.957	.910	.833	.721	.577	.416	.258	.130	.047	.009	.000
	10	1.0	1.0	1.0	1.0	1.0	1.0	.999	.996	.989	.971	.937	.876	.780	.645	.479	.302	.147	.044	.004
	11	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.994	.983	.960	.916	.839	.719	.552	.352	.158	.030
	12	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.997	.992	.979	.953	.899	.802	.643	.415	.153
	13	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.993	.982	.956	.897	.771	.512
15	0	.463	.206	.087	.035	.013	.005	.002	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	1	.829	.549	.319	.167	.080	.035	.014	.005	.002	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	2	.964	.816	.604	.398	.236	.127	.062	.027	.011	.004	.001	.000	.000	.000	.000	.000	.000	.000	.000
	3	.995	.944	.823	.648	.461	.297	.173	.091	.042	.018	.006	.002	.000	.000	.000	.000	.000	.000	.000
	4	.999	.987	.938	.836	.686	.515	.352	.217	.120	.059	.025	.009	.003	.001	.000	.000	.000	.000	.000
	5	1.0	.998	.983	.939	.852	.722	.564	.403	.261	.151	.077	.034	.012	.004	.001	.000	.000	.000	.000
	6	1.0	1.0	.996	.982	.943	.869	.755	.610	.452	.304	.182	.095	.042	.015	.004	.001	.000	.000	.000
	7	1.0	1.0	.999	.996	.983	.950	.887	.787	.654	.500	.346	.213	.113	.050	.017	.004	.001	.000	.000
	8	1.0	1.0	1.0	.999	.996	.985	.958	.905	.818	.696	.548	.390	.245	.131	.057	.018	.004	.000	.000
	9	1.0	1.0	1.0	1.0	.999	.996	.988	.966	.923	.849	.739	.597	.436	.278	.148	.061	.017	.002	.000
	10	1.0	1.0	1.0	1.0	1.0	.999	.997	.991	.975	.941	.880	.783	.648	.485	.314	.164	.062	.013	.001
	11	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.998	.994	.982	.958	.909	.827	.703	.539	.352	.177	.056	.005
	12	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.996	.989	.973	.938	.873	.764	.602	.396	.184	.036
	13	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.998	.995	.986	.965	.920	.833	.681	.451	.171
	14	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.998	.995	.987	.965	.913	.794	.537

Table A2, continued. Binomial distribution

n	x	p																			
		.050	.100	.150	.200	.250	.300	.350	.400	.450	.500	.550	.600	.650	.700	.750	.800	.850	.900	.950	
16	1	.811	.515	.284	.141	.063	.026	.010	.003	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	
	2	.957	.789	.561	.352	.197	.099	.045	.018	.007	.002	.001	.000	.000	.000	.000	.000	.000	.000	.000	
	3	.993	.932	.790	.598	.405	.246	.134	.065	.028	.011	.003	.001	.000	.000	.000	.000	.000	.000	.000	
	4	.999	.983	.921	.798	.630	.450	.289	.167	.085	.038	.015	.005	.001	.000	.000	.000	.000	.000	.000	
	5	1.0	.997	.976	.918	.810	.660	.490	.329	.198	.105	.049	.019	.006	.002	.000	.000	.000	.000	.000	
	6	1.0	.999	.994	.973	.920	.825	.688	.527	.366	.227	.124	.058	.023	.007	.002	.000	.000	.000	.000	
	7	1.0	1.0	.999	.993	.973	.926	.841	.716	.563	.402	.256	.142	.067	.026	.007	.001	.000	.000	.000	
	8	1.0	1.0	1.0	.999	.993	.974	.933	.858	.744	.598	.437	.284	.159	.074	.027	.007	.001	.000	.000	
	9	1.0	1.0	1.0	1.0	.998	.993	.977	.942	.876	.773	.634	.473	.312	.175	.080	.027	.006	.001	.000	
	10	1.0	1.0	1.0	1.0	1.0	.998	.994	.981	.951	.895	.802	.671	.510	.340	.190	.082	.024	.003	.000	
	11	1.0	1.0	1.0	1.0	1.0	1.0	.999	.995	.985	.962	.915	.833	.711	.550	.370	.202	.079	.017	.001	
	12	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.997	.989	.972	.935	.866	.754	.595	.402	.210	.068	.007	
	13	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.998	.993	.982	.955	.901	.803	.648	.439	.211	.043	
	14	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.997	.990	.974	.937	.859	.716	.485	.189	
	15	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.997	.990	.972	.926	.815	.560	
18	1	.774	.450	.224	.099	.039	.014	.005	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	
	2	.942	.734	.480	.271	.135	.060	.024	.008	.003	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	
	3	.989	.902	.720	.501	.306	.165	.078	.033	.012	.004	.001	.000	.000	.000	.000	.000	.000	.000	.000	
	y 4	.998	.972	.879	.716	.519	.333	.189	.094	.041	.015	.005	.001	.000	.000	.000	.000	.000	.000	.000	
	5	1.0	.994	.958	.867	.717	.534	.355	.209	.108	.048	.018	.006	.001	.000	.000	.000	.000	.000	.000	
	6	1.0	.999	.988	.949	.861	.722	.549	.374	.226	.119	.054	.020	.006	.001	.000	.000	.000	.000	.000	
	7	1.0	1.0	.997	.984	.943	.859	.728	.563	.391	.240	.128	.058	.021	.006	.001	.000	.000	.000	.000	
	8	1.0	1.0	.999	.996	.981	.940	.861	.737	.578	.407	.253	.135	.060	.021	.005	.001	.000	.000	.000	
	9	1.0	1.0	1.0	.999	.995	.979	.940	.865	.747	.593	.422	.263	.139	.060	.019	.004	.001	.000	.000	
	10	1.0	1.0	1.0	1.0	.999	.994	.979	.942	.872	.760	.609	.437	.272	.141	.057	.016	.003	.000	.000	
	11	1.0	1.0	1.0	1.0	1.0	.999	.994	.980	.946	.881	.774	.626	.451	.278	.139	.051	.012	.001	.000	
	12	1.0	1.0	1.0	1.0	1.0	1.0	.999	.994	.982	.952	.892	.791	.645	.466	.283	.133	.042	.006	.000	
	13	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.995	.985	.959	.906	.811	.667	.481	.284	.121	.028	.002	
	14	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.996	.988	.967	.922	.835	.694	.499	.280	.098	.011	
	15	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.997	.992	.976	.940	.865	.729	.520	.266	.058
	16	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.995	.986	.961	.901	.776	.550	.226	
20	1	.736	.392	.176	.069	.024	.008	.002	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	
	2	.925	.677	.405	.206	.091	.035	.012	.004	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	
	y 3	.984	.867	.648	.411	.225	.107	.044	.016	.005	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	
	4	.997	.957	.830	.630	.415	.238	.118	.051	.019	.006	.002	.000	.000	.000	.000	.000	.000	.000	.000	
	5	1.0	.989	.933	.804	.617	.416	.245	.126	.055	.021	.006	.002	.000	.000	.000	.000	.000	.000	.000	
	6	1.0	.998	.978	.913	.786	.608	.417	.250	.130	.058	.021	.006	.002	.000	.000	.000	.000	.000	.000	
	7	1.0	1.0	.994	.968	.898	.772	.601	.416	.252	.132	.058	.021	.006	.001	.000	.000	.000	.000	.000	
	8	1.0	1.0	.999	.990	.959	.887	.762	.596	.414	.252	.131	.057	.020	.005	.001	.000	.000	.000	.000	
	9	1.0	1.0	1.0	.997	.986	.952	.878	.755	.591	.412	.249	.128	.053	.017	.004	.001	.000	.000	.000	
	10	1.0	1.0	1.0	.999	.996	.983	.947	.872	.751	.588	.409	.245	.122	.048	.014	.003	.000	.000	.000	
	11	1.0	1.0	1.0	1.0	.999	.995	.980	.943	.869	.748	.586	.404	.238	.113	.041	.010	.001	.000	.000	
	12	1.0	1.0	1.0	1.0	1.0	.999	.994	.979	.942	.868	.748	.584	.399	.228	.102	.032	.006	.000	.000	
	13	1.0	1.0	1.0	1.0	1.0	1.0	.998	.994	.979	.942	.870	.750	.583	.392	.214	.087	.022	.002	.000	
	14	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.998	.994	.979	.945	.874	.755	.584	.383	.196	.067	.011	.000	
	15	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.998	.994	.981	.949	.882	.762	.585	.370	.170	.043	.003
	16	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.995	.984	.956	.893	.775	.589	.352	.133	.016
25	2	.873	.537	.254	.098	.032	.009	.002	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	
	3	.966	.764	.471	.234	.096	.033	.010	.002	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	
	4	.993	.902	.682	.421	.214	.090	.032	.009	.002	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	
	5	.999	.967	.838	.617	.378	.193	.083	.029	.009	.002	.000	.000	.000	.000	.000	.000	.000	.000	.000	
	6	1.0	.991	.930	.780	.561	.341	.173	.074	.026	.007	.002	.000	.000	.000	.000	.000	.000	.000	.000	
	7	1.0	.998	.975	.891	.727	.512	.306	.154	.064	.022	.006	.001	.000	.000	.000	.000	.000	.000	.000	
	8	1.0	1.0	.992	.953	.851	.677	.467	.274	.134	.054	.017	.004	.001	.000	.000	.000	.000	.000	.000	
	9	1.0	1.0	.998	.983	.929	.811	.630	.425	.242	.115	.044	.013	.003	.000	.000	.000	.000	.000	.000	
	10	1.0	1.0	1.0	.994	.970	.902	.771	.586	.384	.212	.096	.034	.009	.002	.000	.000	.000	.000	.000	
	11	1.0	1.0	1.0	.998	.989	.956	.875	.732	.543	.345	.183	.078	.025	.006	.001	.000	.000	.000	.000	
	12	1.0	1.0	1.0	1.0	.997	.983	.940	.846	.694	.500	.306	.154	.060	.017	.003	.000	.000	.000	.000	
	13	1.0	1.0	1.0	1.0	.999	.994	.975	.922	.817	.655	.457	.268	.125	.044	.011	.002	.000	.000	.000	
	14	1.0	1.0	1.0	1.0	1.0	.998	.991	.966	.904	.788	.616	.414	.229	.098	.030	.006	.000	.000	.000	
	15	1.0	1.0	1.0	1.0	1.0	1.0	.997	.987	.956	.885	.758	.575	.370	.189	.071	.017	.002	.000	.000	
	16	1.0	1.0	1.0	1.0	1.0	1.0	.999	.996	.983	.946	.866	.726	.533	.323	.149	.047	.008	.000	.000	
	17	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.999	.994	.978	.936	.846	.694	.488	.273	.109	.025	.002	.000	
	18	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.998	.993	.974	.926	.827	.659	.439	.220	.070	.009	.000	
	19	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.998	.991	.971	.917	.807	.622	.383	.162	.033	.001	
	20	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	.998	.991	.968	.910	.786	.579	.318	.098	.007	

Table A3. Poisson distribution

$$F(x) = P\{X \leq x\} = \sum_{k=0}^x \frac{e^{-\lambda} \lambda^k}{k!}$$

x	λ														
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.1	1.2	1.3	1.4	1.5
0	.905	.819	.741	.670	.607	.549	.497	.449	.407	.368	.333	.301	.273	.247	.223
1	.995	.982	.963	.938	.910	.878	.844	.809	.772	.736	.699	.663	.627	.592	.558
2	1.00	.999	.996	.992	.986	.977	.966	.953	.937	.920	.900	.879	.857	.833	.809
3	1.00	1.00	1.00	.999	.998	.997	.994	.991	.987	.981	.974	.966	.957	.946	.934
4	1.00	1.00	1.00	1.00	1.00	1.00	.999	.999	.998	.996	.995	.992	.989	.986	.981
5	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.999	.998	.998	.997	.996
6	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.999
7	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

x	λ														
	1.6	1.7	1.8	1.9	2.0	2.1	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	3.0
0	.202	.183	.165	.150	.135	.122	.111	.100	.091	.082	.074	.067	.061	.055	.050
1	.525	.493	.463	.434	.406	.380	.355	.331	.308	.287	.267	.249	.231	.215	.199
2	.783	.757	.731	.704	.677	.650	.623	.596	.570	.544	.518	.494	.469	.446	.423
3	.921	.907	.891	.875	.857	.839	.819	.799	.779	.758	.736	.714	.692	.670	.647
4	.976	.970	.964	.956	.947	.938	.928	.916	.904	.891	.877	.863	.848	.832	.815
5	.994	.992	.990	.987	.983	.980	.975	.970	.964	.958	.951	.943	.935	.926	.916
6	.999	.998	.997	.997	.995	.994	.993	.991	.988	.986	.983	.979	.976	.971	.966
7	1.00	1.00	.999	.999	.999	.999	.998	.997	.997	.996	.995	.993	.992	.990	.988
8	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.999	.999	.999	.998	.998	.997	.996
9	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.999	.999	.999
10	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

x	λ														
	3.5	4.0	4.5	5.0	5.5	6.0	6.5	7.0	7.5	8.0	8.5	9.0	9.5	10.0	10.5
0	.030	.018	.011	.007	.004	.002	.002	.001	.001	.000	.000	.000	.000	.000	.000
1	.136	.092	.061	.040	.027	.017	.011	.007	.005	.003	.002	.001	.001	.000	.000
2	.321	.238	.174	.125	.088	.062	.043	.030	.020	.014	.009	.006	.004	.003	.002
3	.537	.433	.342	.265	.202	.151	.112	.082	.059	.042	.030	.021	.015	.010	.007
4	.725	.629	.532	.440	.358	.285	.224	.173	.132	.100	.074	.055	.040	.029	.021
5	.858	.785	.703	.616	.529	.446	.369	.301	.241	.191	.150	.116	.089	.067	.050
6	.935	.889	.831	.762	.686	.606	.527	.450	.378	.313	.256	.207	.165	.130	.102
7	.973	.949	.913	.867	.809	.744	.673	.599	.525	.453	.386	.324	.269	.220	.179
8	.990	.979	.960	.932	.894	.847	.792	.729	.662	.593	.523	.456	.392	.333	.279
9	.997	.992	.983	.968	.946	.916	.877	.830	.776	.717	.653	.587	.522	.458	.397
10	.999	.997	.993	.986	.975	.957	.933	.901	.862	.816	.763	.706	.645	.583	.521
11	1.00	.999	.998	.995	.989	.980	.966	.947	.921	.888	.849	.803	.752	.697	.639
12	1.00	1.00	.999	.998	.996	.991	.984	.973	.957	.936	.909	.876	.836	.792	.742
13	1.00	1.00	1.00	.999	.998	.996	.993	.987	.978	.966	.949	.926	.898	.864	.825
14	1.00	1.00	1.00	1.00	.999	.999	.997	.994	.990	.983	.973	.959	.940	.917	.888
15	1.00	1.00	1.00	1.00	1.00	.999	.999	.998	.995	.992	.986	.978	.967	.951	.932
16	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.998	.996	.993	.989	.982	.973	.960
17	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.998	.997	.995	.991	.986	.978
18	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.999	.998	.996	.993	.988
19	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.999	.998	.997	.994
20	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.998	.997

Table A3, continued. Poisson distribution

x	λ														
	11	12	13	14	15	16	17	18	19	20	22	24	26	28	30
0	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
1	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
2	.001	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
3	.005	.002	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
4	.015	.008	.004	.002	.001	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
5	.038	.020	.011	.006	.003	.001	.001	.000	.000	.000	.000	.000	.000	.000	.000
6	.079	.046	.026	.014	.008	.004	.002	.001	.001	.000	.000	.000	.000	.000	.000
7	.143	.090	.054	.032	.018	.010	.005	.003	.002	.001	.000	.000	.000	.000	.000
8	.232	.155	.100	.062	.037	.022	.013	.007	.004	.002	.001	.000	.000	.000	.000
9	.341	.242	.166	.109	.070	.043	.026	.015	.009	.005	.002	.000	.000	.000	.000
10	.460	.347	.252	.176	.118	.077	.049	.030	.018	.011	.004	.001	.000	.000	.000
11	.579	.462	.353	.260	.185	.127	.085	.055	.035	.021	.008	.003	.001	.000	.000
12	.689	.576	.463	.358	.268	.193	.135	.092	.061	.039	.015	.005	.002	.001	.000
13	.781	.682	.573	.464	.363	.275	.201	.143	.098	.066	.028	.011	.004	.001	.000
14	.854	.772	.675	.570	.466	.368	.281	.208	.150	.105	.048	.020	.008	.003	.001
15	.907	.844	.764	.669	.568	.467	.371	.287	.215	.157	.077	.034	.014	.005	.002
16	.944	.899	.835	.756	.664	.566	.468	.375	.292	.221	.117	.056	.025	.010	.004
17	.968	.937	.890	.827	.749	.659	.564	.469	.378	.297	.169	.087	.041	.018	.007
18	.982	.963	.930	.883	.819	.742	.655	.562	.469	.381	.232	.128	.065	.030	.013
19	.991	.979	.957	.923	.875	.812	.736	.651	.561	.470	.306	.180	.097	.048	.022
20	.995	.988	.975	.952	.917	.868	.805	.731	.647	.559	.387	.243	.139	.073	.035
21	.998	.994	.986	.971	.947	.911	.861	.799	.725	.644	.472	.314	.190	.106	.054
22	.999	.997	.992	.983	.967	.942	.905	.855	.793	.721	.556	.392	.252	.148	.081
23	1.00	.999	.996	.991	.981	.963	.937	.899	.849	.787	.637	.473	.321	.200	.115
24	1.00	.999	.998	.995	.989	.978	.959	.932	.893	.843	.712	.554	.396	.260	.157
25	1.00	1.00	.999	.997	.994	.987	.975	.955	.927	.888	.777	.632	.474	.327	.208
26	1.00	1.00	1.00	.999	.997	.993	.985	.972	.951	.922	.832	.704	.552	.400	.267
27	1.00	1.00	1.00	.999	.998	.996	.991	.983	.969	.948	.877	.768	.627	.475	.333
28	1.00	1.00	1.00	1.00	.999	.998	.995	.990	.980	.966	.913	.823	.697	.550	.403
29	1.00	1.00	1.00	1.00	1.00	.999	.997	.994	.988	.978	.940	.868	.759	.623	.476
30	1.00	1.00	1.00	1.00	1.00	.999	.999	.997	.993	.987	.959	.904	.813	.690	.548
31	1.00	1.00	1.00	1.00	1.00	1.00	.999	.998	.996	.992	.973	.932	.859	.752	.619
32	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.998	.995	.983	.953	.896	.805	.685
33	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.997	.989	.969	.925	.850	.744
34	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.999	.994	.979	.947	.888	.797
35	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.996	.987	.964	.918	.843
36	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.998	.992	.976	.941	.880
37	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.995	.984	.959	.911
38	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.997	.990	.972	.935
39	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.998	.994	.981	.954
40	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.996	.988	.968
41	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.998	.992	.978
42	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.995	.985
43	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.997	.990
44	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.998	.994
45	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.996
46	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999	.998
47	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999
48	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999
49	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	.999
50	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Table A4. Standard Normal distribution

$$\Phi(z) = P\{Z \leq z\} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-x^2/2} dx$$

z	-0.09	-0.08	-0.07	-0.06	-0.05	-0.04	-0.03	-0.02	-0.01	-0.00
-(3.9+)	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
-3.8	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
-3.7	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
-3.6	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0002	.0002
-3.5	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002
-3.4	.0002	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003
-3.3	.0003	.0004	.0004	.0004	.0004	.0004	.0004	.0005	.0005	.0005
-3.2	.0005	.0005	.0005	.0006	.0006	.0006	.0006	.0006	.0007	.0007
-3.1	.0007	.0007	.0008	.0008	.0008	.0008	.0009	.0009	.0009	.0010
-3.0	.0010	.0010	.0011	.0011	.0011	.0012	.0012	.0013	.0013	.0013
-2.9	.0014	.0014	.0015	.0015	.0016	.0016	.0017	.0018	.0018	.0019
-2.8	.0019	.0020	.0021	.0021	.0022	.0023	.0023	.0024	.0025	.0026
-2.7	.0026	.0027	.0028	.0029	.0030	.0031	.0032	.0033	.0034	.0035
-2.6	.0036	.0037	.0038	.0039	.0040	.0041	.0043	.0044	.0045	.0047
-2.5	.0048	.0049	.0051	.0052	.0054	.0055	.0057	.0059	.0060	.0062
-2.4	.0064	.0066	.0068	.0069	.0071	.0073	.0075	.0078	.0080	.0082
-2.3	.0084	.0087	.0089	.0091	.0094	.0096	.0099	.0102	.0104	.0107
-2.2	.0110	.0113	.0116	.0119	.0122	.0125	.0129	.0132	.0136	.0139
-2.1	.0143	.0146	.0150	.0154	.0158	.0162	.0166	.0170	.0174	.0179
-2.0	.0183	.0188	.0192	.0197	.0202	.0207	.0212	.0217	.0222	.0228
-1.9	.0233	.0239	.0244	.0250	.0256	.0262	.0268	.0274	.0281	.0287
-1.8	.0294	.0301	.0307	.0314	.0322	.0329	.0336	.0344	.0351	.0359
-1.7	.0367	.0375	.0384	.0392	.0401	.0409	.0418	.0427	.0436	.0446
-1.6	.0455	.0465	.0475	.0485	.0495	.0505	.0516	.0526	.0537	.0548
-1.5	.0559	.0571	.0582	.0594	.0606	.0618	.0630	.0643	.0655	.0668
-1.4	.0681	.0694	.0708	.0721	.0735	.0749	.0764	.0778	.0793	.0808
-1.3	.0823	.0838	.0853	.0869	.0885	.0901	.0918	.0934	.0951	.0968
-1.2	.0985	.1003	.1020	.1038	.1056	.1075	.1093	.1112	.1131	.1151
-1.1	.1170	.1190	.1210	.1230	.1251	.1271	.1292	.1314	.1335	.1357
-1.0	.1379	.1401	.1423	.1446	.1469	.1492	.1515	.1539	.1562	.1587
-0.9	.1611	.1635	.1660	.1685	.1711	.1736	.1762	.1788	.1814	.1841
-0.8	.1867	.1894	.1922	.1949	.1977	.2005	.2033	.2061	.2090	.2119
-0.7	.2148	.2177	.2206	.2236	.2266	.2296	.2327	.2358	.2389	.2420
-0.6	.2451	.2483	.2514	.2546	.2578	.2611	.2643	.2676	.2709	.2743
-0.5	.2776	.2810	.2843	.2877	.2912	.2946	.2981	.3015	.3050	.3085
-0.4	.3121	.3156	.3192	.3228	.3264	.3300	.3336	.3372	.3409	.3446
-0.3	.3483	.3520	.3557	.3594	.3632	.3669	.3707	.3745	.3783	.3821
-0.2	.3859	.3897	.3936	.3974	.4013	.4052	.4090	.4129	.4168	.4207
-0.1	.4247	.4286	.4325	.4364	.4404	.4443	.4483	.4522	.4562	.4602
-0.0	.4641	.4681	.4721	.4761	.4801	.4840	.4880	.4920	.4960	.5000

Table A4, continued. Standard Normal distribution

[illegible]

Table A5. Table of Student's T-distribution t_α ; critical values, such that $P\{t > t_\alpha\} = \alpha$

ν (d.f.)	α , the right-tail probability									
	.10	.05	.025	.02	.01	.005	.0025	.001	.0005	.0001
1	3.078	6.314	12.706	15.89	31.82	63.66	127.3	318.3	636.6	3185
2	1.886	2.920	4.303	4.849	6.965	9.925	14.09	22.33	31.60	70.71
3	1.638	2.353	3.182	3.482	4.541	5.841	7.453	10.21	12.92	22.20
4	1.533	2.132	2.776	2.999	3.747	4.604	5.598	7.173	8.610	13.04
5	1.476	2.015	2.571	2.757	3.365	4.032	4.773	5.894	6.869	9.676
6	1.440	1.943	2.447	2.612	3.143	3.707	4.317	5.208	5.959	8.023
7	1.415	1.895	2.365	2.517	2.998	3.499	4.029	4.785	5.408	7.064
8	1.397	1.860	2.306	2.449	2.896	3.355	3.833	4.501	5.041	6.442
9	1.383	1.833	2.262	2.398	2.821	3.250	3.690	4.297	4.781	6.009
10	1.372	1.812	2.228	2.359	2.764	3.169	3.581	4.144	4.587	5.694
11	1.363	1.796	2.201	2.328	2.718	3.106	3.497	4.025	4.437	5.453
12	1.356	1.782	2.179	2.303	2.681	3.055	3.428	3.930	4.318	5.263
13	1.350	1.771	2.160	2.282	2.650	3.012	3.372	3.852	4.221	5.111
14	1.345	1.761	2.145	2.264	2.624	2.977	3.326	3.787	4.140	4.985
15	1.341	1.753	2.131	2.249	2.602	2.947	3.286	3.733	4.073	4.880
16	1.337	1.746	2.120	2.235	2.583	2.921	3.252	3.686	4.015	4.790
17	1.333	1.740	2.110	2.224	2.567	2.898	3.222	3.646	3.965	4.715
18	1.330	1.734	2.101	2.214	2.552	2.878	3.197	3.610	3.922	4.648
19	1.328	1.729	2.093	2.205	2.539	2.861	3.174	3.579	3.883	4.590
20	1.325	1.725	2.086	2.197	2.528	2.845	3.153	3.552	3.850	4.539
21	1.323	1.721	2.080	2.189	2.518	2.831	3.135	3.527	3.819	4.492
22	1.321	1.717	2.074	2.183	2.508	2.819	3.119	3.505	3.792	4.452
23	1.319	1.714	2.069	2.177	2.500	2.807	3.104	3.485	3.768	4.416
24	1.318	1.711	2.064	2.172	2.492	2.797	3.091	3.467	3.745	4.382
25	1.316	1.708	2.060	2.167	2.485	2.787	3.078	3.450	3.725	4.352
26	1.315	1.706	2.056	2.162	2.479	2.779	3.067	3.435	3.707	4.324
27	1.314	1.703	2.052	2.158	2.473	2.771	3.057	3.421	3.689	4.299
28	1.313	1.701	2.048	2.154	2.467	2.763	3.047	3.408	3.674	4.276
29	1.311	1.699	2.045	2.150	2.462	2.756	3.038	3.396	3.660	4.254
30	1.310	1.697	2.042	2.147	2.457	2.750	3.030	3.385	3.646	4.234
32	1.309	1.694	2.037	2.141	2.449	2.738	3.015	3.365	3.622	4.198
34	1.307	1.691	2.032	2.136	2.441	2.728	3.002	3.348	3.601	4.168
36	1.306	1.688	2.028	2.131	2.434	2.719	2.990	3.333	3.582	4.140
38	1.304	1.686	2.024	2.127	2.429	2.712	2.980	3.319	3.566	4.115
40	1.303	1.684	2.021	2.123	2.423	2.704	2.971	3.307	3.551	4.094
45	1.301	1.679	2.014	2.115	2.412	2.690	2.952	3.281	3.520	4.049
50	1.299	1.676	2.009	2.109	2.403	2.678	2.937	3.261	3.496	4.014
55	1.297	1.673	2.004	2.104	2.396	2.668	2.925	3.245	3.476	3.985
60	1.296	1.671	2.000	2.099	2.390	2.660	2.915	3.232	3.460	3.962
70	1.294	1.667	1.994	2.093	2.381	2.648	2.899	3.211	3.435	3.926
80	1.292	1.664	1.990	2.088	2.374	2.639	2.887	3.195	3.416	3.899
90	1.291	1.662	1.987	2.084	2.368	2.632	2.878	3.183	3.402	3.878
100	1.290	1.660	1.984	2.081	2.364	2.626	2.871	3.174	3.390	3.861
200	1.286	1.653	1.972	2.067	2.345	2.601	2.838	3.131	3.340	3.789
∞	1.282	1.645	1.960	2.054	2.326	2.576	2.807	3.090	3.290	3.719

Table A5. Table of Student's T-distribution (continued)

v (d.f)	0.45	0.40	0.35	0.30	0.25	0.20	0.15	0.10
1	0,158	0,325	0,51	0,727	1	1,376	1,963	3,078
2	0,142	0,289	0,445	0,617	0,816	1,061	1,386	1,886
3	0,137	0,277	0,424	0,584	0,765	0,978	1,25	1,638
4	0,134	0,271	0,414	0,569	0,741	0,941	1,19	1,533
5	0,132	0,267	0,408	0,559	0,727	0,92	1,156	1,476
6	0,131	0,265	0,404	0,553	0,718	0,906	1,134	1,44
7	0,13	0,263	0,402	0,549	0,711	0,896	1,119	1,415
8	0,13	0,262	0,399	0,546	0,706	0,889	1,108	1,397
9	0,129	0,261	0,398	0,543	0,703	0,883	1,1	1,383
10	0,129	0,26	0,397	0,542	0,7	0,879	1,093	1,372
11	0,129	0,26	0,396	0,54	0,697	0,876	1,088	1,363
12	0,128	0,259	0,395	0,539	0,695	0,873	1,083	1,356
13	0,128	0,259	0,394	0,538	0,694	0,87	1,079	1,35
14	0,128	0,258	0,393	0,537	0,692	0,868	1,076	1,345
15	0,128	0,258	0,393	0,536	0,691	0,866	1,074	1,341
16	0,128	0,258	0,392	0,535	0,69	0,865	1,071	1,337
17	0,128	0,257	0,392	0,534	0,689	0,863	1,069	1,333
18	0,127	0,257	0,392	0,534	0,688	0,862	1,067	1,33
19	0,127	0,257	0,391	0,533	0,688	0,861	1,066	1,328
20	0,127	0,257	0,391	0,533	0,687	0,86	1,064	1,325
21	0,127	0,257	0,391	0,532	0,686	0,859	1,063	1,323
22	0,127	0,256	0,39	0,532	0,686	0,858	1,061	1,321
23	0,127	0,256	0,39	0,532	0,685	0,858	1,06	1,319
24	0,127	0,256	0,39	0,531	0,685	0,857	1,059	1,318
25	0,127	0,256	0,39	0,531	0,684	0,856	1,058	1,316
26	0,127	0,256	0,39	0,531	0,684	0,856	1,058	1,315
27	0,127	0,256	0,389	0,531	0,684	0,855	1,057	1,314
28	0,127	0,256	0,389	0,53	0,683	0,855	1,056	1,313
29	0,127	0,256	0,389	0,53	0,683	0,854	1,055	1,311
30	0,127	0,256	0,389	0,53	0,683	0,854	1,055	1,31

J

NOTE: $P\{t < -t_\alpha\} = P\{t > t_\alpha\}$

Example (d.f=10): $P\{t < -0.7\} = P\{t > 0.7\} = 0.25$

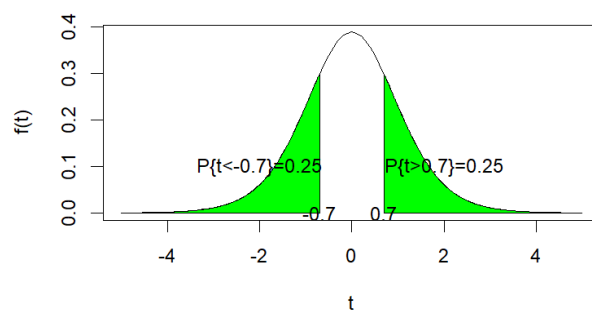


Table A6. Table of Chi-Square Distribution
 χ^2_α ; critical values, such that $P\{\chi^2 > \chi^2_\alpha\} = \alpha$

ν (d.f.)	α , the right-tail probability													
	.999	.995	.99	.975	.95	.90	.80	.20	.10	.05	.025	.01	.005	.001
1	0.00	0.00	0.00	0.00	0.00	0.02	0.06	1.64	2.71	3.84	5.02	6.63	7.88	10.8
2	0.00	0.01	0.02	0.05	0.10	0.21	0.45	3.22	4.61	5.99	7.38	9.21	10.6	13.8
3	0.02	0.07	0.11	0.22	0.35	0.58	1.01	4.64	6.25	7.81	9.35	11.3	12.8	16.3
4	0.09	0.21	0.30	0.48	0.71	1.06	1.65	5.99	7.78	9.49	11.1	13.3	14.9	18.5
5	0.21	0.41	0.55	0.83	1.15	1.61	2.34	7.29	9.24	11.1	12.8	15.1	16.7	20.5
6	0.38	0.68	0.87	1.24	1.64	2.20	3.07	8.56	10.6	12.6	14.4	16.8	18.5	22.5
7	0.60	0.99	1.24	1.69	2.17	2.83	3.82	9.80	12.0	14.1	16.0	18.5	20.3	24.3
8	0.86	1.34	1.65	2.18	2.73	3.49	4.59	11.0	13.4	15.5	17.5	20.1	22.0	26.1
9	1.15	1.73	2.09	2.70	3.33	4.17	5.38	12.2	14.7	16.9	19.0	21.7	23.6	27.9
10	1.48	2.16	2.56	3.25	3.94	4.87	6.18	13.4	16.0	18.3	20.5	23.2	25.2	29.6
11	1.83	2.60	3.05	3.82	4.57	5.58	6.99	14.6	17.3	19.7	21.9	24.7	26.8	31.3
12	2.21	3.07	3.57	4.40	5.23	6.30	7.81	15.8	18.5	21.0	23.3	26.2	28.3	32.9
13	2.62	3.57	4.11	5.01	5.89	7.04	8.63	17.0	19.8	22.4	24.7	27.7	29.8	34.5
14	3.04	4.07	4.66	5.63	6.57	7.79	9.47	18.2	21.1	23.7	26.1	29.1	31.3	36.1
15	3.48	4.60	5.23	6.26	7.26	8.55	10.3	19.3	22.3	25.0	27.5	30.6	32.8	37.7
16	3.94	5.14	5.81	6.91	7.96	9.31	11.1	20.5	23.5	26.3	28.8	32.0	34.3	39.3
17	4.42	5.70	6.41	7.56	8.67	10.1	12.0	21.6	24.8	27.6	30.2	33.4	35.7	40.8
18	4.90	6.26	7.01	8.23	9.39	10.9	12.9	22.8	26.0	28.9	31.5	34.8	37.2	42.3
19	5.41	6.84	7.63	8.91	10.1	11.7	13.7	23.9	27.2	30.1	32.9	36.2	38.6	43.8
20	5.92	7.43	8.26	9.59	10.9	12.4	14.6	25.0	28.4	31.4	34.2	37.6	40.0	45.3
21	6.45	8.03	8.90	10.3	11.6	13.2	15.4	26.2	29.6	32.7	35.5	38.9	41.4	46.8
22	6.98	8.64	9.54	11.0	12.3	14.0	16.3	27.3	30.8	33.9	36.8	40.3	42.8	48.3
23	7.53	9.26	10.2	11.7	13.1	14.8	17.2	28.4	32.0	35.2	38.1	41.6	44.2	49.7
24	8.08	9.89	10.9	12.4	13.8	15.7	18.1	29.6	33.2	36.4	39.4	43.0	45.6	51.2
25	8.65	10.5	11.5	13.1	14.6	16.5	18.9	30.7	34.4	37.7	40.6	44.3	46.9	52.6
26	9.22	11.2	12.2	13.8	15.4	17.3	19.8	31.8	35.6	38.9	41.9	45.6	48.3	54.1
27	9.80	11.8	12.9	14.6	16.2	18.1	20.7	32.9	36.7	40.1	43.2	47.0	49.6	55.5
28	10.4	12.5	13.6	15.3	16.9	18.9	21.6	34.0	37.9	41.3	44.5	48.3	51.0	56.9
29	11.0	13.1	14.3	16.0	17.7	19.8	22.5	35.1	39.1	42.6	45.7	49.6	52.3	58.3
30	11.6	13.8	15.0	16.8	18.5	20.6	23.4	36.3	40.3	43.8	47.0	50.9	53.7	59.7
31	12.2	14.5	15.7	17.5	19.3	21.4	24.3	37.4	41.4	45.0	48.2	52.2	55.0	61.1
32	12.8	15.1	16.4	18.3	20.1	22.3	25.1	38.5	42.6	46.2	49.5	53.5	56.3	62.5
33	13.4	15.8	17.1	19.0	20.9	23.1	26.0	39.6	43.7	47.4	50.7	54.8	57.6	63.9
34	14.1	16.5	17.8	19.8	21.7	24.0	26.9	40.7	44.9	48.6	52.0	56.1	59.0	65.2
35	14.7	17.2	18.5	20.6	22.5	24.8	27.8	41.8	46.1	49.8	53.2	57.3	60.3	66.6
36	15.3	17.9	19.2	21.3	23.3	25.6	28.7	42.9	47.2	51.0	54.4	58.6	61.6	68
37	16.0	18.6	20.0	22.1	24.1	26.5	29.6	44.0	48.4	52.2	55.7	59.9	62.9	69.3
38	16.6	19.3	20.7	22.9	24.9	27.3	30.5	45.1	49.5	53.4	56.9	61.2	64.2	70.7
39	17.3	20.0	21.4	23.7	25.7	28.2	31.4	46.2	50.7	54.6	58.1	62.4	65.5	72.1
40	17.9	20.7	22.2	24.4	26.5	29.1	32.3	47.3	51.8	55.8	59.3	63.7	66.8	73.4
41	18.6	21.4	22.9	25.2	27.3	29.9	33.3	48.4	52.9	56.9	60.6	65.0	68.1	74.7
42	19.2	22.1	23.7	26.0	28.1	30.8	34.2	49.5	54.1	58.1	61.8	66.2	69.3	76.1
43	19.9	22.9	24.4	26.8	29.0	31.6	35.1	50.5	55.2	59.3	63.0	67.5	70.6	77.4
44	20.6	23.6	25.1	27.6	29.8	32.5	36.0	51.6	56.4	60.5	64.2	68.7	71.9	78.7
45	21.3	24.3	25.9	28.4	30.6	33.4	36.9	52.7	57.5	61.7	65.4	70.0	73.2	80.1
46	21.9	25.0	26.7	29.2	31.4	34.2	37.8	53.8	58.6	62.8	66.6	71.2	74.4	81.4
47	22.6	25.8	27.4	30.0	32.3	35.1	38.7	54.9	59.8	64.0	67.8	72.4	75.7	82.7
48	23.3	26.5	28.2	30.8	33.1	35.9	39.6	56.0	60.9	65.2	69.0	73.7	77.0	84.0
49	24.0	27.2	28.9	31.6	33.9	36.8	40.5	57.1	62.0	66.3	70.2	74.9	78.2	85.4
50	24.7	28.0	29.7	32.4	34.8	37.7	41.4	58.2	63.2	67.5	71.4	76.2	79.5	86.7

Table A8. Table of Critical Values for the Wilcoxon Signed Rank Test

For the left-tail test, the table gives the largest integer w such that $\mathbf{P}\{W \leq w \mid H_0\} \leq \alpha$.

For the right-tail test, the table gives the smallest integer w such that $\mathbf{P}\{W \geq w \mid H_0\} \leq \alpha$.

A missing table entry means that such an integer does not exist among possible values of W .

n	α , left-tail probability for the left-tail test							α , right-tail probability for the right-tail test						
	0.001	0.005	0.010	0.025	0.050	0.100	0.200	0.200	0.100	0.050	0.025	0.010	0.005	0.001
1	—	—	—	—	—	—	—	—	—	—	—	—	—	—
2	—	—	—	—	—	—	—	—	—	—	—	—	—	—
3	—	—	—	—	—	—	0	6	—	—	—	—	—	—
4	—	—	—	—	—	0	2	8	10	—	—	—	—	—
5	—	—	—	—	0	2	3	12	13	15	—	—	—	—
6	—	—	—	0	2	3	5	16	18	19	21	—	—	—
7	—	—	0	2	3	5	8	20	23	25	26	28	—	—
8	—	0	1	3	5	8	11	25	28	31	33	35	36	—
9	—	1	3	5	8	10	14	31	35	37	40	42	44	—
10	0	3	5	8	10	14	18	37	41	45	47	50	52	55
11	1	5	7	10	13	17	22	44	49	53	56	59	61	65
12	2	7	9	13	17	21	27	51	57	61	65	69	71	76
13	4	9	12	17	21	26	32	59	65	70	74	79	82	87
14	6	12	15	21	25	31	38	67	74	80	84	90	93	99
15	8	15	19	25	30	36	44	76	84	90	95	101	105	112
16	11	19	23	29	35	42	50	86	94	101	107	113	117	125
17	14	23	27	34	41	48	57	96	105	112	119	126	130	139
18	18	27	32	40	47	55	65	106	116	124	131	139	144	153
19	21	32	37	46	53	62	73	117	128	137	144	153	158	169
20	26	37	43	52	60	69	81	129	141	150	158	167	173	184
21	30	42	49	58	67	77	90	141	154	164	173	182	189	201
22	35	48	55	65	75	86	99	154	167	178	188	198	205	218
23	40	54	62	73	83	94	109	167	182	193	203	214	222	236
24	45	61	69	81	91	104	119	181	196	209	219	231	239	255
25	51	68	76	89	100	113	130	195	212	225	236	249	257	274
26	58	75	84	98	110	124	141	210	227	241	253	267	276	293
27	64	83	92	107	119	134	153	225	244	259	271	286	295	314
28	71	91	101	116	130	145	165	241	261	276	290	305	315	335
29	79	100	110	126	140	157	177	258	278	295	309	325	335	356
30	86	109	120	137	151	169	190	275	296	314	328	345	356	379

Table A9. Critical Values for the Mann-Whitney-Wilcoxon Rank-Sum Test

For the left-tail test, the table gives the largest integer u such that $\mathbf{P}\{U \leq u \mid H_0\} \leq \alpha$.

For the right-tail test, the table gives the smallest integer u such that $\mathbf{P}\{U \geq u \mid H_0\} \leq \alpha$.

A missing table entry means that such an integer does not exist among possible values of U .

n_1	n_2	α , left-tail probability for the left-tail test $H_A : X$ is stochastically smaller than Y							α , right-tail probability for the right-tail test $H_A : X$ is stochastically larger than Y						
		0.001	0.005	0.010	0.025	0.050	0.100	0.200	0.200	0.100	0.050	0.025	0.010	0.005	0.001
3	2	—	—	—	—	—	6	7	11	12	—	—	—	—	—
3	3	—	—	—	—	6	7	8	13	14	15	—	—	—	—
3	4	—	—	—	—	6	7	9	15	17	18	—	—	—	—
3	5	—	—	—	6	7	8	10	17	19	20	21	—	—	—
3	6	—	—	—	7	8	9	11	19	21	22	23	—	—	—
3	7	—	—	6	7	8	10	12	21	23	25	26	27	—	—
3	8	—	—	6	8	9	11	13	23	25	27	28	30	—	—
3	9	—	6	7	8	10	11	14	25	28	29	31	32	33	—
3	10	—	6	7	9	10	12	15	27	30	32	33	35	36	—
3	11	—	6	7	9	11	13	16	29	32	34	36	38	39	—
3	12	—	7	8	10	11	14	17	31	34	37	38	40	41	—
4	2	—	—	—	—	—	10	11	17	18	—	—	—	—	—
4	3	—	—	—	—	10	11	13	19	21	22	—	—	—	—
4	4	—	—	—	10	11	13	14	22	23	25	26	—	—	—
4	5	—	—	10	11	12	14	15	25	26	28	29	30	—	—
4	6	—	10	11	12	13	15	17	27	29	31	32	33	34	—
4	7	—	10	11	13	14	16	18	30	32	34	35	37	38	—
4	8	—	11	12	14	15	17	20	32	35	37	38	40	41	—
4	9	—	11	13	14	16	19	21	35	37	40	42	43	45	—
4	10	10	12	13	15	17	20	23	37	40	43	45	47	48	50
4	11	10	12	14	16	18	21	24	40	43	46	48	50	52	54
4	12	10	13	15	17	19	22	26	42	46	49	51	53	55	58
5	2	—	—	—	—	15	16	17	23	24	25	—	—	—	—
5	3	—	—	—	15	16	17	19	26	28	29	30	—	—	—
5	4	—	—	15	16	17	19	20	30	31	33	34	35	—	—
5	5	—	15	16	17	19	20	22	33	35	36	38	39	40	—
5	6	—	16	17	18	20	22	24	36	38	40	42	43	44	—
5	7	—	16	18	20	21	23	26	39	42	44	45	47	49	—
5	8	15	17	19	21	23	25	28	42	45	47	49	51	53	55
5	9	16	18	20	22	24	27	30	45	48	51	53	55	57	59
5	10	16	19	21	23	26	28	32	48	52	54	57	59	61	64
5	11	17	20	22	24	27	30	34	51	55	58	61	63	65	68
5	12	17	21	23	26	28	32	36	54	58	62	64	67	69	73
6	2	—	—	—	—	21	22	23	31	32	33	—	—	—	—
6	3	—	—	—	22	23	24	26	34	36	37	38	—	—	—
6	4	—	21	22	23	24	26	28	38	40	42	43	44	45	—
6	5	—	22	23	24	26	28	30	42	44	46	48	49	50	—
6	6	—	23	24	26	28	30	33	45	48	50	52	54	55	—
6	7	21	24	25	27	29	32	35	49	52	55	57	59	60	63
6	8	22	25	27	29	31	34	37	53	56	59	61	63	65	68
6	9	23	26	28	31	33	36	40	56	60	63	65	68	70	73
6	10	24	27	29	32	35	38	42	60	64	67	70	73	75	78
6	11	25	28	30	34	37	40	44	64	68	71	74	78	80	83
6	12	25	30	32	35	38	42	47	67	72	76	79	82	84	89
7	2	—	—	—	—	28	29	31	39	41	42	—	—	—	—
7	3	—	—	28	29	30	32	34	43	45	47	48	49	—	—
7	4	—	28	29	31	32	34	36	48	50	52	53	55	56	—
7	5	—	29	31	33	34	36	39	52	55	57	58	60	62	—
7	6	28	31	32	34	36	39	42	56	59	62	64	66	67	70
7	7	29	32	34	36	39	41	45	60	64	66	69	71	73	76
7	8	30	34	35	38	41	44	48	64	68	71	74	77	78	82
7	9	31	35	37	40	43	46	50	69	73	76	79	82	84	88
7	10	33	37	39	42	45	49	53	73	77	81	84	87	89	93
7	11	34	38	40	44	47	51	56	77	82	86	89	93	95	99
7	12	35	40	42	46	49	54	59	81	86	91	94	98	100	105

Table A9, continued. Critical Values for the Mann-Whitney-Wilcoxon Rank-Sum Test

For the left-tail test, the table gives the largest integer u such that $P\{U \leq u \mid H_0\} \leq \alpha$.

For the right-tail test, the table gives the smallest integer u such that $P\{U \geq u \mid H_0\} \leq \alpha$.

A missing table entry means that such an integer does not exist among possible values of U .

n_1	n_2	α , left-tail probability for the left-tail test $H_A : X$ is stochastically smaller than Y							α , right-tail probability for the right-tail test $H_A : X$ is stochastically larger than Y						
		0.001	0.005	0.010	0.025	0.050	0.100	0.200	0.200	0.100	0.050	0.025	0.010	0.005	0.001
8	2	—	—	—	36	37	38	40	48	50	51	52	—	—	—
8	3	—	—	36	38	39	41	43	53	55	57	58	60	—	—
8	4	—	37	38	40	41	43	46	58	61	63	64	66	67	—
8	5	36	38	40	42	44	46	49	63	66	68	70	72	74	76
8	6	37	40	42	44	46	49	52	68	71	74	76	78	80	83
8	7	38	42	43	46	49	52	56	72	76	79	82	85	86	90
8	8	40	43	45	49	51	55	59	77	81	85	87	91	93	96
8	9	41	45	47	51	54	58	62	82	86	90	93	97	99	103
8	10	42	47	49	53	56	60	65	87	92	96	99	103	105	110
8	11	44	49	51	55	59	63	69	91	97	101	105	109	111	116
8	12	45	51	53	58	62	66	72	96	102	106	110	115	117	123
9	2	—	—	—	45	46	47	49	59	61	62	63	—	—	—
9	3	—	45	46	47	49	50	53	64	67	68	70	71	72	—
9	4	—	46	48	49	51	54	56	70	72	75	77	78	80	—
9	5	46	48	50	52	54	57	60	75	78	81	83	85	87	89
9	6	47	50	52	55	57	60	64	80	84	87	89	92	94	97
9	7	48	52	54	57	60	63	67	86	90	93	96	99	101	105
9	8	50	54	56	60	63	67	71	91	95	99	102	106	108	112
9	9	52	56	59	62	66	70	75	96	101	105	109	112	115	119
9	10	53	58	61	65	69	73	78	102	107	111	115	119	122	127
9	11	55	61	63	68	72	76	82	107	113	117	121	126	128	134
9	12	57	63	66	71	75	80	86	112	118	123	127	132	135	141
10	2	—	—	—	55	56	58	60	70	72	74	75	—	—	—
10	3	—	55	56	58	59	61	64	76	79	81	82	84	85	—
10	4	55	57	58	60	62	65	68	82	85	88	90	92	93	95
10	5	56	59	61	63	66	68	72	88	92	94	97	99	101	104
10	6	58	61	63	66	69	72	76	94	98	101	104	107	109	112
10	7	60	64	66	69	72	76	80	100	104	108	111	114	116	120
10	8	61	66	68	72	75	79	84	106	111	115	118	122	124	129
10	9	63	68	71	75	79	83	88	112	117	121	125	129	132	137
10	10	65	71	74	78	82	87	93	117	123	128	132	136	139	145
10	11	67	73	77	81	86	91	97	123	129	134	139	143	147	153
10	12	69	76	79	84	89	94	101	129	136	141	146	151	154	161
11	2	—	—	—	66	67	69	71	83	85	87	88	—	—	—
11	3	—	66	67	69	71	73	76	89	92	94	96	98	99	—
11	4	66	68	70	72	74	77	80	96	99	102	104	106	108	110
11	5	68	71	73	75	78	81	85	102	106	109	112	114	116	119
11	6	70	73	75	79	82	85	89	109	113	116	119	123	125	128
11	7	72	76	78	82	85	89	94	115	120	124	127	131	133	137
11	8	74	79	81	85	89	93	99	121	127	131	135	139	141	146
11	9	76	82	84	89	93	97	103	128	134	138	142	147	149	155
11	10	78	84	88	92	97	102	108	134	140	145	150	154	158	164
11	11	81	87	91	96	100	106	112	141	147	153	157	162	166	172
11	12	83	90	94	99	104	110	117	147	154	160	165	170	174	181
12	2	—	—	—	79	80	82	84	96	98	100	101	—	—	—
12	3	—	79	80	82	83	86	89	103	106	109	110	112	113	—
12	4	78	81	83	85	87	90	94	110	114	117	119	121	123	126
12	5	80	84	86	89	91	95	99	117	121	125	127	130	132	136
12	6	82	87	89	92	95	99	104	124	129	133	136	139	141	146
12	7	85	90	92	96	99	104	109	131	136	141	144	148	150	155
12	8	87	93	95	100	104	108	114	138	144	148	152	157	159	165
12	9	90	96	99	104	108	113	119	145	151	156	160	165	168	174
12	10	92	99	102	107	112	117	124	152	159	164	169	174	177	184
12	11	95	102	106	111	116	122	129	159	166	172	177	182	186	193
12	12	98	105	109	115	120	127	134	166	173	180	185	191	195	202

Medidas estadísticas

Medida	Poblacional	Muestral
	<i>Población ordenada</i> = $(x_1, x_2, \dots, x_N)$	<i>Muestra ordenada</i> = $(X_1, X_2, \dots, X_n)$
Tamaño (número de elementos)	N	n
Media (aritmética)	$\mu = \frac{\sum_{i=1}^N x_i}{N}$	$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$
Moda	Valor x_i que más se repite	Valor X_i que más se repite
Cuantil p	Si $p \cdot N$ no es un número entero: $q_p = x_{[p \cdot N] + 1}$ Si $p \cdot N$ es un número entero: $q_p = \frac{x_{p \cdot N} + x_{p \cdot N + 1}}{2}$ Donde $[p \cdot N]$ = parte entera de $p \cdot N$	Si $p \cdot n$ no es un número entero: $\hat{q}_p = x_{[p \cdot n] + 1}$ Si $p \cdot n$ es un número entero: $\hat{q}_p = \frac{X_{p \cdot n} + X_{p \cdot n + 1}}{2}$ Donde $[p \cdot n]$ = parte entera de $p \cdot n$
Valor mínimo, valor máximo, rango	$\min x = x_1, \max x = x_N$ rango $x = x_N - x_1$	$\min X = X_1, \max X = X_n$ rango $X = X_n - X_1$
Varianza	$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$	También llamada cuasivarianza muestral $s^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}$
Desviación estándar	$\sigma = \sqrt{\sigma^2}$	$s = \sqrt{s^2}$
Coefficiente de variación	$cv = \frac{\sigma}{\mu}$	$CV = \frac{s}{\bar{X}}$
Rango intercuartílico	$IQR = Q_3 - Q_1$ Regla datos atípicos: $[Q_1 - 1.5 \cdot IQR, Q_3 + 1.5 \cdot IQR]$	$I\hat{Q}R = \hat{Q}_3 - \hat{Q}_1$ Regla datos atípicos: $[\hat{Q}_1 - 1.5 \cdot I\hat{Q}R, \hat{Q}_3 + 1.5 \cdot I\hat{Q}R]$
Sesgo o coeficiente de asimetría	$A = \frac{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^3}{\sigma^3}$	$\hat{A} = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^3}{s^3}$
Curtosis o coeficiente de apuntamiento	$K = \frac{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^4}{\sigma^4} - 3$	$\hat{K} = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^4}{s^4} - 3$
Proporción de elementos que cumplen una condición	$p = \frac{n^{\circ} \text{ elementos que cumplen la condición}}{N}$	$\hat{p} = \frac{n^{\circ} \text{ elementos que cumplen la condición}}{n}$

Medidas estadísticas a partir de una tabla de frecuencias

Fila	x ó X	f	h	F	H
1	v_1	f_1	h_1	F_1	H_1
2	v_2	f_2	h_2	F_2	H_2
...	...	...	...	...	...
i	v_i	f_i	h_i	F_i	H_i
...	...	...	...	...	...
k	v_k	f_k	h_k	F_k	H_k

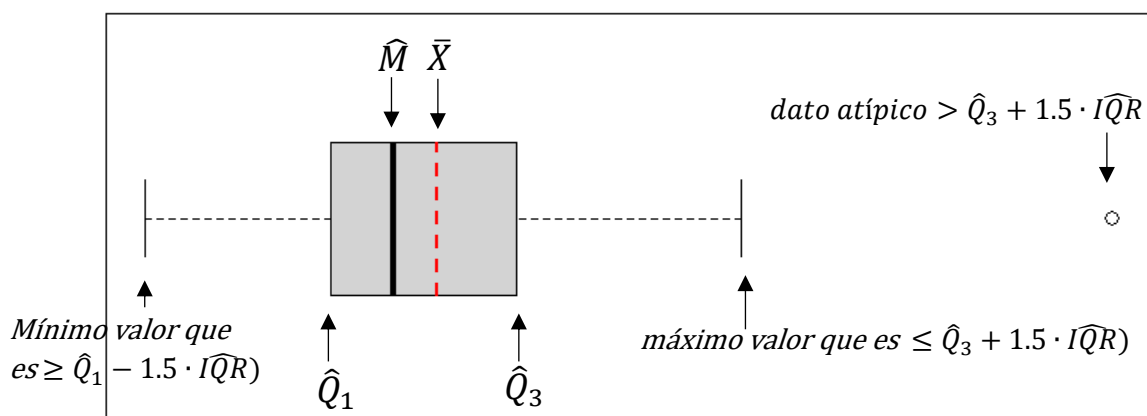
Fila	x ó X	Marca	f	h	F	H
1	$[a_1, b_1)$	m_1	f_1	h_1	F_1	H_1
2	$[a_2, b_2)$	m_2	f_2	h_2	F_2	H_2
...	...	...	...	...	...	...
i	$[a_i, b_i)$	m_i	f_i	h_i	F_i	H_i
...	...	...	...	...	...	...
k	$[a_k, b_k]$	m_k	f_k	h_k	F_k	H_k

Regla de Sturges: $k = \lceil 1 + 3,322 \log_{10} N \rceil$ (población) ó $k = \lceil 1 + 3,322 \log_{10} n \rceil$ (muestra)

Ejemplos: $10 \leq n \leq 15 \rightarrow k = 5$; $16 \leq n \leq 31 \rightarrow k = 6$; $32 \leq n \leq 63 \rightarrow k = 7$; $64 \leq n \leq 127 \rightarrow k = 8$

Medida	Poblacional	Muestral
Media (aritmética)	$\mu = \frac{\sum_{i=1}^k f_i v_i}{N}$	$\bar{X} = \frac{\sum_{i=1}^k f_i v_i}{n}$
Media (aritmética) con datos agrupados	$\mu = \frac{\sum_{i=1}^k f_i m_i}{N}$ $m_i = \frac{a_i + b_i}{2}$	$\bar{X} = \frac{\sum_{i=1}^k f_i m_i}{n}$
Varianza	$\sigma^2 = \frac{\sum_{i=1}^k f_i (v_i - \mu)^2}{N}$	$s^2 = \frac{\sum_{i=1}^k f_i (v_i - \bar{X})^2}{n - 1}$
Varianza con datos agrupados	$\sigma^2 = \frac{\sum_{i=1}^k f_i (m_i - \mu)^2}{N}$	$s^2 = \frac{\sum_{i=1}^k f_i (m_i - \bar{X})^2}{n - 1}$
Cuantil	Siendo v_i el primer valor que cumple: $F_i \geq p \cdot N$ Si $F_i > p \cdot N$: $q_p = v_i$ Si $F_i = p \cdot N$: $q_p = \frac{v_i + v_{i+1}}{2}$	Siendo v_i el primer valor que: $F_i \geq p \cdot n$ Si $F_i > p \cdot n$: $\hat{q}_p = v_i$ Si $F_i = p \cdot n$: $\hat{q}_p = \frac{v_i + v_{i+1}}{2}$
Cuantil con datos agrupados	Siendo $[a_i, b_i)$ el primero que cumple: $F_i \geq p \cdot N$ $q_p = a_i + (b_i - a_i) \cdot \frac{p \cdot N - F_{i-1}}{f_i}$	Siendo $[a_i, b_i)$ el primero que: $F_i \geq p \cdot n$ $\hat{q}_p = a_i + (b_i - a_i) \cdot \frac{p \cdot n - F_{i-1}}{f_i}$

Diagrama de caja para una muestra



Estadística bidimensional

<i>Medida</i>	<i>Poblacional</i>	<i>Muestral</i>
<i>Tamaño</i>	N $\text{Población} = ((x_1, y_1), (x_2, y_2), \dots, (x_N, y_N))$	n $\text{Muestra} = ((X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n))$
Medias (aritméticas) marginales	$\mu_x = \frac{\sum_{i=1}^N x_i}{N}$	$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$
	$\mu_y = \frac{\sum_{i=1}^N y_i}{N}$	$\bar{Y} = \frac{\sum_{i=1}^n Y_i}{n}$
Varianzas marginales	$\sigma_x^2 = \frac{\sum_{i=1}^N (x_i - \mu_x)^2}{N}$	$s_X^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}$
	$\sigma_y^2 = \frac{\sum_{i=1}^N (y_i - \mu_y)^2}{N}$	$s_Y^2 = \frac{\sum_{i=1}^n (Y_i - \bar{Y})^2}{n - 1}$
Covarianza	$\sigma_{xy} = \frac{\sum_{i=1}^N (x_i - \mu_x)(y_i - \mu_y)}{N}$	$s_{XY} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$
Coeficiente de correlación (de Pearson)	$\rho = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$	$r = \frac{s_{XY}}{s_X s_Y}$

Medidas estadísticas a partir de una tabla de contingencia sin datos agrupados

	w_1	w_2	...	w_j	...	w_l	f_X	h_X
v_1	$f_{1,1}$	$f_{1,2}$	...	$f_{1,j}$	...	$f_{1,l}$	f_{v_1}	h_{v_1}
v_2	$f_{2,1}$	$f_{2,2}$	...	$f_{2,j}$	...	$f_{2,l}$	f_{v_2}	h_{v_2}
...	...	...	...	...	...	...	...	...
v_i	$f_{i,1}$	$f_{i,2}$	...	$f_{i,j}$	...	$f_{i,l}$	f_{v_i}	h_{v_i}
...	...	...	...	...	...	...	...	...
v_k	$f_{k,1}$	$f_{k,2}$	...	$f_{k,j}$	...	$f_{k,l}$	f_{v_k}	h_v
f_Y	f_{w_1}	f_{w_2}	...	f_{w_j}	...	f_{w_l}	N ó n	
h_Y	h_{w_1}	h_{w_2}	...	h_{w_j}	...	h_{w_l}		1

Medida	Poblacional	Muestral
Media (aritmética) marginal de x (ó X)	$\mu_x = \frac{\sum_{i=1}^k f_{v_i} v_i}{N}$	$\bar{X} = \frac{\sum_{i=1}^k f_{v_i} v_i}{n}$
Varianza marginal de x (ó X)	$\sigma_x^2 = \frac{\sum_{i=1}^k f_{v_i} (v_i - \mu_x)^2}{N}$	$s_X^2 = \frac{\sum_{i=1}^k f_{v_i} (v_i - \bar{X})^2}{n-1}$
Cuantil marginal de x (ó X)	Siendo v_i el primer valor que: $F_{v_i} \geq p_x \cdot N$ Si $F_{v_i} > p_x \cdot N$: $q_{p_x} = v_i$ Si $F_{v_i} = p_x \cdot N$: $q_{p_x} = \frac{v_i + v_{i+1}}{2}$ Donde $F_{v_i} = \sum_{r=1}^i f_{v_r}$	Siendo v_i el primer valor que: $F_{v_i} \geq p_x \cdot n$ Si $F_{v_i} > p_x \cdot n$: $\hat{q}_{p_x} = v_i$ Si $F_{v_i} = p_x \cdot n$: $\hat{q}_{p_x} = \frac{v_i + v_{i+1}}{2}$ Donde $F_{v_i} = \sum_{r=1}^i f_{v_r}$
Media condicionada de x (ó X) cuando y (ó Y) = w_j	$\mu_x w_j = \frac{\sum_{i=1}^k f_{i,j} v_i}{f_{w_j}}$	$\bar{X} w_j = \frac{\sum_{i=1}^k f_{i,j} v_i}{f_{w_j}}$
Varianza condicionada de x (ó X) cuando y (ó Y) = w_j	$\sigma_x^2 w_j = \frac{\sum_{i=1}^k f_{i,j} (v_i - \mu_x w_j)^2}{f_{w_j}}$	$s_X^2 w_j = \frac{\sum_{i=1}^k f_{i,j} (v_i - \bar{X} w_j)^2}{f_{w_j} - 1}$
Cuantil condicionado p de x (ó X) cuando y (ó Y) = w_j	Siendo v_i el primer valor que: $F_{i,j} \geq p_x \cdot f_{w_j}$ Si $F_{i,j} > p_x \cdot f_{w_j}$: $q_{p_x} w_j = v_i$ Si $F_{i,j} = p_x \cdot f_{w_j}$: $q_{p_x} w_j = \frac{v_i + v_{i+1}}{2}$ Donde $F_{i,j} = \sum_{r=1}^i f_{r,j}$	Siendo v_i el primer valor que: $F_{i,j} \geq p_x \cdot f_{w_j}$ Si $F_{i,j} > p_x \cdot f_{w_j}$: $\hat{q}_{p_x} w_j = v_i$ Si $F_{i,j} = p_x \cdot f_{w_j}$: $\hat{q}_{p_x} w_j = \frac{v_i + v_{i+1}}{2}$ Donde $F_{i,j} = \sum_{r=1}^i f_{r,j}$
Independencia	$\frac{f_{i,j}}{N} = h_{v_i} \cdot h_{w_j} \text{ para todo } i \text{ y } j$	$\frac{f_{i,j}}{n} = h_{v_i} \cdot h_{w_j} \text{ para todo } i \text{ y } j$

Medidas estadísticas a partir de una tabla de contingencia con datos agrupados

	w_1	w_2	...	w_j	...	w_l	f_X	h_X
$c_1 = [a_1, b_1)$ m_1	$f_{1,1}$	$f_{1,2}$	...	$f_{1,j}$	...	$f_{1,l}$	f_{c_1}	h_{c_1}
$c_2 = [a_2, b_2)$ m_2	$f_{2,1}$	$f_{2,2}$	...	$f_{2,j}$	...	$f_{2,l}$	f_{c_2}	h_{c_2}
...	...	...	...	...	...	...	...	...
$c_i = [a_i, b_i)$ m_i	$f_{i,1}$	$f_{i,2}$	...	$f_{i,j}$	...	$f_{i,l}$	f_{c_i}	h_{c_i}
...	...	...	...	...	...	...	...	...
$c_k = [a_k, b_k]$ m_k	$f_{k,1}$	$f_{k,2}$	...	$f_{k,j}$	...	$f_{k,l}$	f_{c_k}	h_{c_k}
f_Y	f_{w_1}	f_{w_2}	...	f_{w_j}	...	f_{w_l}	N ó n	
h_Y	h_{Y_1}	h_{Y_2}	...	h_{Y_j}	...	h_{Y_l}		1

Medida	Poblacional	Muestral
Media (aritmética) marginal de x (ó X)	$\mu_x = \frac{\sum_{i=1}^k f_{c_i} m_i}{N}$	$\bar{X} = \frac{\sum_{i=1}^k f_{c_i} m_i}{n}$
Varianza marginal de x (ó X)	$\sigma_x^2 = \frac{\sum_{i=1}^k f_{c_i} (m_i - \mu_x)^2}{N}$	$s_x^2 = \frac{\sum_{i=1}^k f_{c_i} (m_i - \bar{X})^2}{n - 1}$
Cuantil marginal p de x (ó X)	Siendo $[a_i, b_i)$ el primero que cumple: $F_{c_i} \geq p_x \cdot N$ $q_{p_x} = a_i + (b_i - a_i) \cdot \frac{p_x \cdot N - F_{c_{i-1}}}{f_{c_i}}$ Donde $F_{c_i} = \sum_{r=1}^i f_{c_r}$	Siendo $[a_i, b_i)$ el primero que cumple: $F_{c_i} \geq p_x \cdot n$ $\hat{q}_{p_x} = a_i + (b_i - a_i) \cdot \frac{p_x \cdot n - F_{c_{i-1}}}{f_{c_i}}$ Donde $F_{c_i} = \sum_{r=1}^i f_{c_r}$
Media condicionada de x (ó X) cuando y (ó Y) = w_j	$\mu_x w_j = \frac{\sum_{i=1}^k f_{i,j} m_i}{f_{w_j}}$	$\bar{X} w_j = \frac{\sum_{i=1}^k f_{i,j} m_i}{f_{w_j}}$
Varianza condicionada de x (ó X) cuando y (ó Y) = w_j	$\sigma_x^2 w_j = \frac{\sum_{i=1}^k f_{i,j} (m_i - \mu_x w_j)^2}{f_{w_j}}$	$s_x^2 w_j = \frac{\sum_{i=1}^k f_{i,j} (m_i - \bar{X} w_j)^2}{f_{w_j} - 1}$
Cuantil condicionado p de x (ó X) cuando y (ó Y) = w_j	Siendo $[a_i, b_i)$ el primero que cumple: $F_{i,j} \geq p_x \cdot f_{w_j}$ $q_{p_x} w_j = a_i + (b_i - a_i) \cdot \frac{p_x \cdot f_{w_j} - F_{i-1,j}}{f_{i,j}}$ Donde $F_{i,j} = \sum_{r=1}^i f_{r,j}$	Siendo $[a_i, b_i)$ el primero que cumple: $F_{i,j} \geq p_x \cdot f_{w_j}$ $\hat{q}_{p_x} w_j = a_i + (b_i - a_i) \cdot \frac{p_x \cdot f_{w_j} - F_{i-1,j}}{f_{i,j}}$ Donde $F_{i,j} = \sum_{r=1}^i f_{r,j}$
Independencia	$\frac{f_{i,j}}{N} = h_{c_i} \cdot h_{w_j} \text{ para todo } i \text{ y } j$	$\frac{f_{i,j}}{n} = h_{c_i} \cdot h_{w_j} \text{ para todo } i \text{ y } j$

COMBINATORIA

Operación	Se pueden repetir los elementos	Fórmula
PERMUTACIONES o VARIACIONES	Si	$P_r(n, k) = n \cdot n \cdots n = n^k$
	No	$P(n, k) = n(n-1)(n-2) \cdots (n-k+1) = \frac{n!}{(n-k)!}$
COMBINACIONES	Si	$C_r(n, k) = \binom{k+n-1}{k} = \frac{(k+n-1)!}{k!(n-1)!}$
	No	$C(n, k) = \binom{n}{k} = \frac{n!}{k!(n-k)!}$

$n! = \text{factorial del número entero } n = 1 \cdot 2 \cdot 3 \cdot 4 \cdots n$, siendo $0! = 1$

PROBABILIDADES

$$P\{A \cap B\} = P\{A\} \cdot P\{B\} \text{ si } A \text{ y } B \text{ son independientes}$$

$$P\{A \cup B\} = P\{A\} + P\{B\} - P\{A \cap B\}$$

$$P\{\bar{A}\} = 1 - P\{A\}$$

$$P\{A \setminus B\} = P\{A - B\} = P\{A\} - P\{A \cap B\}$$

$$P\{A|B\} = \frac{P\{A \cap B\}}{P\{B\}}$$

$$P\{B|A\} = \frac{P\{A|B\} \cdot P\{B\}}{P\{A\}} = \frac{P\{A|B\} \cdot P\{B\}}{P\{A|B\} \cdot P\{B\} + P\{A|\bar{B}\} \cdot P\{\bar{B}\}}$$

$$P\{A\} = \sum_{j=1}^k P\{A|B_j\} \cdot P\{B_j\}$$

$$\text{Si } P\{B_i \cap B_j\} = 0 \text{ para cualquier } i \neq j$$

$$\text{y si } \sum_{j=1}^k P\{B_j\} = 1$$

VARIABLES ALEATORIAS

$$\mu = E(X) = \sum_x xP(x)$$

$$E(Y) = E\{g(X)\} = \sum_x g(x)P(x)$$

$$\sigma^2 = Var(X) = E(X - E(X))^2 = \sum_x (x - \mu)^2 P(x) = E(X^2) - \mu^2$$

$$P\{|X - \mu| > \varepsilon\} \leq \left(\frac{\sigma}{\varepsilon}\right)^2$$

$$P(X + Y) = P(X) + P(Y)$$

$$P_{(X,Y)}(x, y) = P\{(X, Y) = (x, y)\} = P\{\{X = x\} \cap \{Y = y\}\}$$

$$P_X(x) = P\{X = x\} = \sum_y P_{(X,Y)}(x, y)$$

$$P_Y(y) = P\{Y = y\} = \sum_x P_{(X,Y)}(x, y)$$

$$P_{(X,Y)}(x, y) = P_X(x)P_Y(y) \text{ si } X \text{ e } Y \text{ son independientes}$$

$$E\{g(X, Y)\} = \sum_x \sum_y g(x, y)P_{(X,Y)}(x, y)$$

$$E(X + Y) = E(X) + E(Y)$$

$$E(X \cdot Y) = E(X) \cdot E(Y) \text{ si } X \text{ e } Y \text{ son independientes}$$

Distribución discreta	Descripción	Función de masa de probabilidad (pmf) $P(x)$	Función de distribución acumulada (cdf) $F(x) = P\{X \leq x\}$	$E(X)$	$Var(X)$
X: H(N, n, p) Hipergeometría	<i>X = Número de éxitos (elementos que cumplen una condición) en n ensayos no independientes de Bernoulli, que consisten en extraer sin reposición n elementos de un conjunto inicial de N elementos</i> <i>n = número de ensayos (extracciones) = 1, 2, 3, ..., N</i> <i>p = probabilidad de éxito en el primer ensayo = proporción de elementos que cumplen la condición en el conjunto inicial</i> <i>x=0, 1, 2, ..., n</i> <i>Si N es mucho más grande que n: $H(N, n, p) \approx \text{Binomial}(n, p)$ (Se puede aceptar con $N > 10n$)</i>	$\frac{\binom{Np}{x} \binom{N(1-p)}{n-x}}{\binom{N}{n}}$	$\sum_{i=0}^x \frac{\binom{Np}{i} \binom{N(1-p)}{n-i}}{\binom{N}{n}}$	np	$np(1-p) \frac{(N-n)}{(N-1)}$

Distribución discreta	Descripción	Función de masa de probabilidad (pmf) $P(x)$	Función de distribución acumulada (cdf) $F(x) = P\{X \leq x\}$	$E(X)$	$Var(X)$
X: Bernoulli(p)	$X = \text{Número de éxitos en un ensayo de Bernoulli}$ $p = \text{probabilidad de éxito}$ $x = 0, 1$	$\begin{cases} 1 - p, & x = 0 \\ p, & x = 1 \end{cases}$	$\begin{cases} 1 - p, & x = 0 \\ 1, & x = 1 \end{cases}$	p	$p(1 - p)$
X: Binomial(n, p)	$X = \text{Número de éxitos en } n \text{ ensayos independientes de Bernoulli}$ $n = \text{número de ensayos} = 1, 2, 3, \dots$ $p = \text{probabilidad de éxito en un ensayo}$ $x = 0, 1, 2, \dots, n$ <i>Si n es grande:</i> $\text{Binomial}(n, p) \approx \text{Poisson}(np)$ $\text{Binomial}(n, p) \approx \text{Normal}(np, \sqrt{np(1 - p)})$ (Aplicado una corrección de continuidad)	$\binom{n}{x} p^x (1 - p)^{n-x}$	$\sum_{i=0}^x \binom{n}{i} p^i (1 - p)^{n-i}$ <i>Tabla A2</i>	np	$np(1 - p)$
X: Geometric(p)	$X = \text{Número de ensayos de Bernoulli hasta el primer éxito}$ $p = \text{probabilidad de éxito en un ensayo}$ $x = 1, 2, 3, \dots$	$p(1 - p)^{x-1}$	$1 - (1 - p)^x$	$\frac{1}{p}$	$\frac{1 - p}{p^2}$
X: NB(k, p) <i>Binomial negativa</i>	$X = \text{Número de ensayos de Bernoulli hasta el éxito } k\text{-ésimo}$ $k = \text{número de éxitos} = 1, 2, 3, \dots$ $p = \text{probabilidad de éxito en un ensayo}$ $x = k, (k+1), (k+2), \dots$	$\binom{x-1}{k-1} p^k (1 - p)^{x-k}$	$\sum_{i=k}^x \binom{i-1}{k-1} p^k (1 - p)^{i-k}$	$\frac{k}{p}$	$\frac{k(1 - p)}{p^2}$
X: Poisson(λ)	$X = \text{Número de "eventos raros" durante un intervalo de tiempo fijo}$ $\lambda = \text{frecuencia de los "eventos raros"} \in (0, \infty)$ $x = 1, 2, 3, \dots$	$e^{-\lambda} \frac{\lambda^x}{x!}$ <i>Siendo $e \approx 2.72$</i>	$\sum_{i=1}^x e^{-\lambda} \frac{\lambda^i}{i!}$ <i>Tabla A3</i>	λ	λ

Distribución continua	Descripción	Función de densidad de probabilidad (pdf) $f(x)$	Función de distribución acumulada (cdf) $F(x) = P(X \leq x)$	$E(X)$	$Var(X)$
X: Uniform(a, b)	$X = \text{Número elegido "al azar" en el intervalo } (a, b)$ $a < x < b$	$\frac{1}{b-a}$	$\frac{1}{b-a}x - \frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
X: Exponential(λ)	$X = \text{Tiempo entre dos eventos con distribución de Poisson}$ $\lambda = \text{frecuencia de ocurrencia de eventos} > 0$ $x > 0$	$\lambda e^{-\lambda x}$	$1 - e^{-\lambda x}$ <i>Siendo $e \approx 2.72$</i>	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
X: Gamma(α, λ)	$X = \text{Tiempo entre dos secuencias de } \alpha \text{ eventos con distribución de Poisson}$ $\alpha = \text{número de eventos en una secuencia} > 0$ $\lambda = \text{frecuencia de ocurrencia de eventos} > 0$ $x > 0$	$\frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$ $\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt$	$\int_{-\infty}^x \frac{\lambda^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\lambda t} dt$ $F(x) \approx P\{Y \geq \alpha\}$ <i>siendo $Y: \text{Poisson}(\lambda x)$</i>	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$
X: Normal(μ, σ)	$\mu = \text{esperanza} \in (-\infty, +\infty)$ $\sigma = \text{desviación estándar} \in (0, +\infty)$ $-\infty < x < \infty$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$\int_{-\infty}^x \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(t-\mu)^2}{2\sigma^2}} dt$	μ	σ^2
Z: Normal(0,1)	$Z = \text{Variable Normal estándar}$ $\text{Si } X: N(\mu, \sigma) \rightarrow Z = \frac{X - \mu}{\sigma}$ $\text{Si } S_n = \sum_{i=1}^n X_i \text{ con } n \text{ grande} \rightarrow Z = \frac{S_n - n\mu}{\sigma\sqrt{n}}$	$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$	$\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$ <i>Tabla A4</i>	0	1

Intervalos de confianza (se asume una distribución Normal de los datos)

Parámetro	Varianza conocida	Estimador	Límite inferior	Límite superior
μ	Si	$Z = \frac{\bar{X} - \bar{x}}{\sigma/\sqrt{n}}$	$L_i = \bar{x} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$	$L_s = \bar{x} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$
	No $n \geq 30$	$Z = \frac{\bar{X} - \bar{x}}{s/\sqrt{n}}$	$L_i = \bar{x} - z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$	$L_s = \bar{x} + z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}$
	No $n < 30$ Distribución Normal	$t_{n-1} = \frac{\bar{X} - \bar{x}}{s/\sqrt{n}}$	$L_i = \bar{x} - t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}}$	$L_s = \bar{x} + t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}}$
p	-	$Z = \frac{\hat{P} - \hat{p}}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}}$	$L_i = \hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$	$L_s = \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
σ^2	No	$\chi^2(n-1)$	$L_i = \frac{(n-1)s^2}{\chi_{\frac{\alpha}{2}, n-1}^2}$	$L_s = \frac{(n-1)s^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2}$

Siendo:

$$z_{\frac{\alpha}{2}} = \text{cuantil} \left(1 - \frac{\alpha}{2} \right) \text{ de } Z: N(0,1) \rightarrow P \left\{ Z < z_{\frac{\alpha}{2}} \right\} = 1 - \frac{\alpha}{2} \quad (\text{Tabla A4})$$

$$t_{\frac{\alpha}{2}, n-1} = \text{cuantil} \left(1 - \frac{\alpha}{2} \right) \text{ de } T: \text{Student}(n-1) \rightarrow P \left\{ T < t_{\frac{\alpha}{2}, n-1} \right\} = 1 - \frac{\alpha}{2} \rightarrow P \left\{ T > t_{\frac{\alpha}{2}, n-1} \right\} = \frac{\alpha}{2} \quad (\text{Tabla A5})$$

$$\chi_{\frac{\alpha}{2}, n-1}^2 = \text{cuantil} \left(1 - \frac{\alpha}{2} \right) \text{ de } \chi^2: \text{Chi-cuadrado}(n-1) \rightarrow P \left\{ \chi^2 < \chi_{\frac{\alpha}{2}, n-1}^2 \right\} = 1 - \frac{\alpha}{2} \rightarrow P \left\{ \chi^2 > \chi_{\frac{\alpha}{2}, n-1}^2 \right\} = \frac{\alpha}{2} \quad (\text{Tabla A6})$$

$$\chi_{1-\frac{\alpha}{2}, n-1}^2 = \text{cuantil} \left(\frac{\alpha}{2} \right) \text{ de } \chi^2: \text{Chi-cuadrado}(n-1) \rightarrow P \left\{ \chi^2 < \chi_{1-\frac{\alpha}{2}, n-1}^2 \right\} = \frac{\alpha}{2} \rightarrow P \left\{ \chi^2 > \chi_{1-\frac{\alpha}{2}, n-1}^2 \right\} = 1 - \frac{\alpha}{2} \quad (\text{Tabla A6})$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Intervalos de confianza (se asume una distribución Normal de los datos)

Diferencia de parámetros	Varianzas conocidas	Límite inferior	Límite superior
$\mu_X - \mu_Y$	Si	$L_i = \bar{x} - \bar{y} - z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$	$L_s = \bar{x} - \bar{y} + z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}$
	No $n, m \geq 30$	$L_i = \bar{x} - \bar{y} - z_{\frac{\alpha}{2}} \sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}$	$L_s = \bar{x} - \bar{y} + z_{\frac{\alpha}{2}} \sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}$
	No $n, m < 30$ $\sigma_X = \sigma_Y$	$L_i = \bar{x} - \bar{y} - \left(t_{\frac{\alpha}{2}, n+m-2}\right)(s_p) \sqrt{\frac{1}{n} + \frac{1}{m}}$	$L_s = \bar{x} - \bar{y} + \left(t_{\frac{\alpha}{2}, n+m-2}\right)(s_p) \sqrt{\frac{1}{n} + \frac{1}{m}}$
	No $n, m < 30$ $\sigma_X \neq \sigma_Y$	$L_i = \bar{x} - \bar{y} - t_{\frac{\alpha}{2}, v} \sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}$	$L_s = \bar{x} - \bar{y} + t_{\frac{\alpha}{2}, v} \sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}$
	Muestras pareadas $n \geq 30$	$L_i = \bar{d} - z_{\frac{\alpha}{2}} \frac{s_D}{\sqrt{n}}$	$L_s = \bar{d} + z_{\frac{\alpha}{2}} \frac{s_D}{\sqrt{n}}$
	Muestras pareadas $n < 30$	$L_i = \bar{d} - t_{\frac{\alpha}{2}, n-1} \frac{s_D}{\sqrt{n}}$	$L_s = \bar{d} + t_{\frac{\alpha}{2}, n-1} \frac{s_D}{\sqrt{n}}$
$p_1 - p_2$	$n, m \geq 30$	$L_i = \hat{p}_1 - \hat{p}_2 - z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$	$L_s = \hat{p}_1 - \hat{p}_2 + z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$

Siendo:

$$s_p = \sqrt{\frac{(n-1)s_X^2 + (m-1)s_Y^2}{n+m-2}}$$

$$v = \frac{\left(\frac{s_X^2}{n} + \frac{s_Y^2}{m}\right)^2}{\frac{s_X^4}{n^2(n-1)} + \frac{s_Y^4}{m^2(m-1)}}$$

$$D = X - Y$$

$$s_X^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$s_Y^2 = \frac{1}{m-1} \sum_{i=1}^m (y_i - \bar{y})^2$$

Contrastes paramétricos (se asume una distribución Normal de los datos)

Parámetro	Varianza conocida	Estadístico de contraste	Contraste de hipótesis	Región de aceptación de H_0	p-valor
μ	Si	$Z_o = \frac{\bar{x} - \mu_o}{\sigma/\sqrt{n}}$	$H_0: \mu = \mu_o$ vs $H_A: \mu < \mu_o$	$(-z_{\alpha}, +\infty)$	$P\{Z \leq Z_o\}$
			$H_0: \mu = \mu_o$ vs $H_A: \mu > \mu_o$	$(-\infty, z_{\alpha})$	$P\{Z \geq Z_o\}$
			$H_0: \mu = \mu_o$ vs $H_A: \mu \neq \mu_o$	$(-z_{\alpha/2}, z_{\alpha/2})$	$P\{ Z \geq Z_o \} = 2P\{Z \geq Z_o \}$
	No $n \geq 30$	$Z_o = \frac{\bar{x} - \mu_o}{s/\sqrt{n}}$	$H_0: \mu = \mu_o$ vs $H_A: \mu < \mu_o$	$(-z_{\alpha}, +\infty)$	$P\{Z \leq Z_o\}$
			$H_0: \mu = \mu_o$ vs $H_A: \mu > \mu_o$	$(-\infty, z_{\alpha})$	$P\{Z \geq Z_o\}$
			$H_0: \mu = \mu_o$ vs $H_A: \mu \neq \mu_o$	$(-z_{\alpha/2}, z_{\alpha/2})$	$P\{ Z \geq Z_o \} = 2P\{Z \geq Z_o \}$
	No $n < 30$	$t_o = \frac{\bar{x} - \mu_o}{s/\sqrt{n}}$	$H_0: \mu = \mu_o$ vs $H_A: \mu < \mu_o$	$(-t_{\alpha, n-1}, +\infty)$	$P\{t \leq t_o\}$
			$H_0: \mu = \mu_o$ vs $H_A: \mu > \mu_o$	$(-\infty, t_{\alpha, n-1})$	$P\{t \geq t_o\}$
			$H_0: \mu = \mu_o$ vs $H_A: \mu \neq \mu_o$	$(-t_{\alpha/2, n-1}, t_{\alpha/2, n-1})$	$P\{ t \geq t_o \} = 2P\{t \geq t_o \}$
p	-	$Z_o = \frac{\hat{p} - p_o}{\sqrt{\frac{p_o(1-p_o)}{n}}}$	$H_0: p = p_o$ vs $H_A: p < p_o$	$(-z_{\alpha}, +\infty)$	$P\{Z \leq Z_o\}$
			$H_0: p = p_o$ vs $H_A: p > p_o$	$(-\infty, z_{\alpha})$	$P\{Z \geq Z_o\}$
			$H_0: p = p_o$ vs $H_A: p \neq p_o$	$(-z_{\alpha/2}, z_{\alpha/2})$	$P\{ Z \geq Z_o \} = 2P\{Z \geq Z_o \}$

Siendo:

$$z_{\frac{\alpha}{2}} = \text{cuantil}\left(1 - \frac{\alpha}{2}\right) \text{ de } Z: N(0,1) \rightarrow P\left\{Z \leq z_{\frac{\alpha}{2}}\right\} = 1 - \frac{\alpha}{2} \quad (\text{Tabla A4})$$

$$z_{\alpha} = \text{cuantil}(1 - \alpha) \text{ de } Z: N(0,1) \rightarrow P\{Z \leq z_{\alpha}\} = 1 - \alpha \quad (\text{Tabla A4})$$

$$t_{\frac{\alpha}{2}, n-1} = \text{cuantil}\left(1 - \frac{\alpha}{2}\right) \text{ de } T: \text{Student}(n-1) \rightarrow P\left\{T \leq t_{\frac{\alpha}{2}, n-1}\right\} = 1 - \frac{\alpha}{2} \rightarrow P\left\{T > t_{\frac{\alpha}{2}, n-1}\right\} = \frac{\alpha}{2} \quad (\text{Tabla A5})$$

$$t_{\alpha, n-1} = \text{cuantil}(1 - \alpha) \text{ de } T: \text{Student}(n-1) \rightarrow P\{T \leq t_{\alpha, n-1}\} = 1 - \alpha \rightarrow P\{T > t_{\alpha, n-1}\} = \alpha \quad (\text{Tabla A5})$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Contrastes paramétricos (se asume una distribución Normal de los datos)

Diferencia de parámetros	Varianzas conocidas	Estadístico de contraste	Contraste de hipótesis	Región de aceptación de H_0	p-valor
$\mu_X - \mu_Y$ Muestras independientes	Si	$Z_o = \frac{\bar{x} - \bar{y} - D}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}}$	$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y < D$	$(-z_{\alpha}, +\infty)$	$P\{Z \leq Z_o\}$
			$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y > D$	$(-\infty, z_{\alpha})$	$P\{Z \geq Z_o\}$
			$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y \neq D$	$(-z_{\alpha/2}, z_{\alpha/2})$	$P\{ Z \geq Z_o \} = 2P\{Z \geq Z_o \}$
	No $n, m \geq 30$ $\sigma_X \neq \sigma_Y$	$Z_o = \frac{\bar{x} - \bar{y} - D}{\sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}}$	$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y < D$	$(-z_{\alpha}, +\infty)$	$P\{Z \leq Z_o\}$
			$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y > D$	$(-\infty, z_{\alpha})$	$P\{Z \geq Z_o\}$
			$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y \neq D$	$(-z_{\alpha/2}, z_{\alpha/2})$	$P\{ Z \geq Z_o \} = 2P\{Z \geq Z_o \}$
	No $n, m \geq 30$ $\sigma_X = \sigma_Y$	$Z_o = \frac{\bar{x} - \bar{y} - D}{s_p \sqrt{\frac{1}{n} + \frac{1}{m}}}$	$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y < D$	$(-z_{\alpha}, +\infty)$	$P\{Z \leq Z_o\}$
			$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y > D$	$(-\infty, z_{\alpha})$	$P\{Z \geq Z_o\}$
			$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y \neq D$	$(-z_{\alpha/2}, z_{\alpha/2})$	$P\{ Z \geq Z_o \} = 2P\{Z \geq Z_o \}$
	No $n, m < 30$ $\sigma_X \neq \sigma_Y$	$t_o = \frac{\bar{x} - \bar{y} - D}{\sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}}$	$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y < D$	$(-t_{\alpha, v}, +\infty)$	$P\{t \leq t_o\}$
			$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y > D$	$(-\infty, t_{\alpha, v})$	$P\{t \geq t_o\}$
			$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y \neq D$	$(-t_{\alpha/2, v}, t_{\alpha/2, v})$	$P\{ t \geq t_o \} = 2P\{t \geq t_o \}$
	No $n, m < 30$ $\sigma_X = \sigma_Y$	$t_o = \frac{\bar{x} - \bar{y} - D}{s_p \sqrt{\frac{1}{n} + \frac{1}{m}}}$	$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y < D$	$(-t_{\alpha, n+m-2}, +\infty)$	$P\{t \leq t_o\}$
			$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y > D$	$(-\infty, t_{\alpha, n+m-2})$	$P\{t \geq t_o\}$
			$H_0: \mu_X - \mu_Y = D$ vs $H_A: \mu_X - \mu_Y \neq D$	$(-t_{\alpha/2, n+m-2}, t_{\alpha/2, n+m-2})$	$P\{ t \geq t_o \} = 2P\{t \geq t_o \}$
$\mu_{(X-Y)}$ Muestras pareadas	No $n \geq 30$	$Z_o = \frac{(\bar{X} - \bar{Y}) - D}{s_{(X-Y)}/\sqrt{n}}$	$H_0: \mu_{(X-Y)} = D$ vs $H_A: \mu_{(X-Y)} < D$	$(-z_{\alpha}, +\infty)$	$P\{Z \leq Z_o\}$
			$H_0: \mu_{(X-Y)} = D$ vs $H_A: \mu_{(X-Y)} > D$	$(-\infty, z_{\alpha})$	$P\{Z \geq Z_o\}$
			$H_0: \mu_{(X-Y)} = D$ vs $H_A: \mu_{(X-Y)} \neq D$	$(-z_{\alpha/2}, z_{\alpha/2})$	$P\{ Z \geq Z_o \} = 2P\{Z \geq Z_o \}$
	No $n < 30$	$t_o = \frac{(\bar{X} - \bar{Y}) - D}{s_{(X-Y)}/\sqrt{n}}$	$H_0: \mu_{(X-Y)} = D$ vs $H_A: \mu_{(X-Y)} < D$	$(-t_{\alpha, n-1}, +\infty)$	$P\{t \leq t_o\}$
			$H_0: \mu_{(X-Y)} = D$ vs $H_A: \mu_{(X-Y)} > D$	$(-\infty, t_{\alpha, n-1})$	$P\{t \geq t_o\}$
			$H_0: \mu_{(X-Y)} = D$ vs $H_A: \mu_{(X-Y)} \neq D$	$(-t_{\alpha/2, n-1}, t_{\alpha/2, n-1})$	$P\{ t \geq t_o \} = 2P\{t \geq t_o \}$
$p_1 - p_2$ Muestras independientes	-	$Z_o = \frac{\hat{p}_1 - \hat{p}_2 - D}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n} + \frac{\hat{p}_2(1-\hat{p}_2)}{m}}}$	$H_0: p_1 - p_2 = D$ vs $H_A: p_1 - p_2 < D$	$(-z_{\alpha}, +\infty)$	$P\{Z \leq Z_o\}$
			$H_0: p_1 - p_2 = D$ vs $H_A: p_1 - p_2 > D$	$(-\infty, z_{\alpha})$	$P\{Z \geq Z_o\}$
			$H_0: p_1 - p_2 = D$ vs $H_A: p_1 - p_2 \neq D$	$(-z_{\alpha/2}, z_{\alpha/2})$	$P\{ Z \geq Z_o \} = 2P\{Z \geq Z_o \}$

Siendo: $s_p = \sqrt{\frac{(n-1)s_X^2 + (m-1)s_Y^2}{n+m-2}}$

$$v = \frac{\left(\frac{s_X^2}{n} + \frac{s_Y^2}{m}\right)^2}{\frac{s_X^4}{n^2(n-1)} + \frac{s_Y^4}{m^2(m-1)}}$$

$$s_X^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$s_Y^2 = \frac{1}{m-1} \sum_{i=1}^m (y_i - \bar{y})^2$$

Contrastes no paramétricos (se asume una distribución desconocida de los datos)

Parámetro	Tipo de contraste	Estadístico de contraste	Valor observado del estadístico de contraste	Contraste de hipótesis	p-valor
M	Sign	Si $n < 30$: Tabla A2 $S = B\left(n, \frac{1}{2}\right)$ Si $n \geq 30$: $S \approx N\left(\frac{n}{2}, \frac{\sqrt{n}}{2}\right)$	S_o = número de observaciones $> m_o$	$H_0: M = m_o$ vs $H_A: M > m_o$	$P\{S \geq S_o\}$
				$H_0: M = m_o$ vs $H_A: M < m_o$	$P\{S \leq S_o\}$
				$H_0: M = m_o$ vs $H_A: M \neq m_o$	$2 \cdot \min(P\{S \leq S_o\}, P\{S \geq S_o\})$
M	Wilcoxon signed rank	Si $n < 15$: Tabla A8 Si $n \geq 15$: $W \approx N\left(\frac{n(n+1)}{4}, \sqrt{\frac{n(n+1)(2n+1)}{24}}\right)$	$W_o = \sum_{i: X_i > m_o} R_i$	$H_0: M = m_o$ vs $H_A: M > m_o$	$P\{W \geq W_o\}$
				$H_0: M = m_o$ vs $H_A: M < m_o$	$P\{W \leq W_o\}$
				$H_0: M = m_o$ vs $H_A: M \neq m_o$	$2 \cdot \min(P\{W \leq W_o\}, P\{W \geq W_o\})$
$M_{(X-Y)}$ Muestras pareadas, con mismo tamaño n	Wilcoxon signed rank paired	Si $n < 15$: Tabla A8 Si $n \geq 15$: $W \approx N\left(\frac{n(n+1)}{4}, \sqrt{\frac{n(n+1)(2n+1)}{24}}\right)$	$W_o = \sum_{i: D_i > 0} R_i$	$H_0: M_{(X-Y)} = 0$ vs $H_A: M_{(X-Y)} > 0$	$P\{W \geq W_o\}$
				$H_0: M_{(X-Y)} = 0$ vs $H_A: M_{(X-Y)} < 0$	$P\{W \leq W_o\}$
				$H_0: M_{(X-Y)} = 0$ vs $H_A: M_{(X-Y)} \neq 0$	$2 \cdot \min(P\{W \leq W_o\}, P\{W \geq W_o\})$
$M_X - M_Y$ Muestras independientes con tamaños n y m	Mann-Whitney-Wilcoxon rank sum	Si n, m $< 10 \rightarrow$ Tabla A9 Si n, m ≥ 10 : $U \approx N\left(\frac{n(n+m+1)}{2}, \sqrt{\frac{n \cdot m(n+m+1)}{12}}\right)$	$U_o = \sum_{i: X_i} R_i$	$H_0: M_X - M_Y = 0$ vs $H_A: M_X - M_Y > 0$	$P\{U \geq U_o\}$
				$H_0: M_X - M_Y = 0$ vs $H_A: M_X - M_Y < 0$	$P\{U \leq U_o\}$
				$H_0: M_X - M_Y = 0$ vs $H_A: M_X - M_Y \neq 0$	$2 \cdot \min(P\{U \leq U_o\}, P\{U \geq U_o\})$

En las aproximaciones Normales hay que aplicar la corrección de continuidad: $P\{X < x\} = P\{X < x - 0.5\}$ ó $P\{X \leq x\} = P\{X < x + 0.5\}$

Regresión y correlación

$$\hat{y} = b_0 + b_1 x \quad b_1 = r \frac{s_y}{s_x} \quad r = \frac{s_{xy}}{s_x s_y} \quad s_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1} \quad b_0 = \bar{y} - b_1 \bar{x}$$

$$s_x^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

Intervalo de confianza de una predicción

Límite inferior	Límite superior
$L_i = \hat{y}_o - \left(t_{\frac{\alpha}{2}, n-2}\right) \cdot (s) \cdot \sqrt{1 + \frac{1}{n} + \frac{(x_o - \bar{x})^2}{S_{xx}}}$	$L_s = \hat{y}_o + \left(t_{\frac{\alpha}{2}, n-2}\right) \cdot (s) \cdot \sqrt{1 + \frac{1}{n} + \frac{(x_o - \bar{x})^2}{S_{xx}}}$

Siendo:

$$\hat{y}_o = (\hat{y}|x = x_o)$$

$$s = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n-2}} = \sqrt{\frac{(1-r^2) \cdot s_y^2 \cdot (n-1)}{n-2}}$$

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$$

$$t_{\frac{\alpha}{2}, n-2} = \text{cuantil} \left(1 - \frac{\alpha}{2}\right) \text{ de T: Student}(n-2)$$

$$P\left\{T < t_{\frac{\alpha}{2}, n-2}\right\} = 1 - \frac{\alpha}{2} \rightarrow P\left\{T > t_{\frac{\alpha}{2}, n-2}\right\} = \frac{\alpha}{2} \quad (\text{Tabla A5})$$